Matrices

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Definition 0.1 (Matrix)**.** A **matrix** [1] of dimensions $m \times n$ is a rectangular array of $m \cdot n$ numbers, arranged in m rows and n columns:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Remark (Matrix entries, and vectors as matrices)

Remark 0.1 (Matrix entries, and vectors as matrices). The entry in row i and column j of $\mathbf{A}$ is written a_{ij} or $(\mathbf{A})_{ij}$. For example, for the 2×3 matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix},$$

$a_{12} = 2$ and $a_{23} = 6$.
A column vector of length p is a $p \times 1$ matrix, and a row vector of length p is a $1 \times p$ matrix. For example, the column vector $(7, 8)$ is a 2×1 matrix, and the row vector $[7,\ 8]$ is a 1×2 matrix.

Definition 0.2 (Equal matrices)**.** Two matrices $\mathbf{A}$ and $\mathbf{B}$ are **equal**, written $\mathbf{A} = \mathbf{B}$, if both are $m \times n$ and $a_{ij} = b_{ij}$ for every $i = 1, \ldots, m$ and $j = 1, \ldots, n$.

Example 0.1 (Equal and unequal matrices)**.**

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

The second pair differs in one entry. The third pair lists the same four numbers in the

same order, but one matrix is 1×4 and the other 2×2, so the two are not equal.

Exercise 0.1 (Solving for an unknown entry)**.** Two matrices are equal only when every pair of matching entries agrees, so one matrix equation stands for several equations between numbers. Let

$$\begin{bmatrix} 2 & x+1 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 0 & 5 \end{bmatrix}.$$

Find x.

Hint

Both matrices are 2×2, so by Definition 0.2 each entry on the left is the same number as the entry in the same position on the right.

Solution

Solution 0.1. Both matrices are 2×2, so they are equal when their entries in each position agree. Three positions already agree, and the entries in position $(1, 2)$ give

$$\begin{aligned} x + 1 &= 4 \quad \text{(Definition 0.2)} \\ x &= 3. \quad \text{(subtract 1 from both sides)} \end{aligned}$$

Definition 0.3 (Row of a matrix)**.** The ith **row** of an $m \times n$ matrix $\mathbf{A}$ is the $1 \times n$ row vector of the entries in row i:

$$\mathbf{A}_{i,:} = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}.$$

Definition 0.4 (Column of a matrix)**.** The jth **column** of an $m \times n$ matrix $\mathbf{A}$ is the column vector of the entries in column j:

$$\mathbf{A}_{:,j} = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

Example 0.2 (Rows and columns of a matrix)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, the rows are

$\mathbf{A}_{1,:} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ and $\mathbf{A}_{2,:} = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$, and the columns are $\mathbf{A}_{:,1} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $\mathbf{A}_{:,2} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ and $\mathbf{A}_{:,3} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$.

Exercise 0.2 (Reading off a column of a matrix)**.** When each row of a matrix records one observation, each column holds one measured quantity for every observation, so reading off a column extracts that quantity's values. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 3 & 4 \end{bmatrix}.$$

Find $\mathbf{A}_{:,2}$, the second column of $\mathbf{A}$.

Hint

The jth column of an $m \times n$ matrix lists the entries $a_{1j}, \ldots, a_{mj}$ from top to bottom, as in Example 0.2.

Solution

Solution 0.2. $\mathbf{A}$ has $m = 2$ rows, so its second column holds the two entries a_{12} and a_{22}:

$$\mathbf{A}_{:,2} = \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

Exercise 0.3 (Counting the rows and columns of a matrix)**.** Data with one row per observation and one column per measured quantity is stored as a matrix, so the matrix's number of rows is the number of observations and its number of columns is the number of quantities. Let

$$\mathbf{C} = \begin{bmatrix} 4 & 0 \\ 1 & -3 \\ 2 & 7 \end{bmatrix}.$$

Find m and n such that $\mathbf{C}$ is an $m \times n$ matrix.

Hint

In Definition 0.1, m counts the rows and n counts the columns.

Solution

Solution 0.3. $\mathbf{C}$ has three rows, $\begin{bmatrix} 4 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & -3 \end{bmatrix}$ and $\begin{bmatrix} 2 & 7 \end{bmatrix}$, and each row has two entries, so $\mathbf{C}$ has two columns. So $m = 3$ and $n = 2$: $\mathbf{C}$ is a 3×2 matrix, with $3 \cdot 2 = 6$ entries.

Exercise 0.4 (Reading off a row of a matrix)**.** When each row of a matrix records one observation, reading off a row extracts every measured quantity for that one observation. Let

$$\mathbf{D} = \begin{bmatrix} 1 & -4 & 2 \\ 0 & 6 & 3 \\ 5 & 5 & -1 \end{bmatrix}.$$

Find $\mathbf{D}_{3,:}$, the third row of $\mathbf{D}$.

Hint

A row of a matrix lists the entries $a_{i1}, \ldots, a_{in}$ from left to right, as in Example 0.2.

Solution

Solution 0.4. Write d_{ij} for the entry of $\mathbf{D}$ in row i and column j. $\mathbf{D}$ has $n = 3$ columns, so its third row holds the three entries d_{31}, d_{32} and d_{33}:

$$\mathbf{D}_{3,:} = \begin{bmatrix} d_{31} & d_{32} & d_{33} \end{bmatrix} = \begin{bmatrix} 5 & 5 & -1 \end{bmatrix}.$$

Theorem 0.1 (A matrix is a column of its rows)**.** *An $m \times n$ matrix $\mathbf{A}$ is the column vector whose entries are its rows:*

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{1,:} \\ \vdots \\ \mathbf{A}_{m,:} \end{bmatrix}.$$

Exercise 0.5 (Stacking rows into a matrix)**.** Theorem 0.1 says that a matrix is determined by its rows, so data recorded one observation at a time can be assembled into a matrix by stacking those records. Let $\mathbf{M}$ be the 3×2 matrix with

$$\mathbf{M}_{1,:} = \begin{bmatrix} 1 & 2 \end{bmatrix}, \qquad \mathbf{M}_{2,:} = \begin{bmatrix} 3 & -1 \end{bmatrix}, \qquad \mathbf{M}_{3,:} = \begin{bmatrix} 0 & 4 \end{bmatrix}.$$

What is $\mathbf{M}_{:,2}$, the second column of $\mathbf{M}$?

Hint

Stack the rows from top to bottom, as in Theorem 0.1, then read down the second column.

Solution

Solution 0.5. Stacking the three rows from top to bottom gives

$$\begin{aligned}\mathbf{M} &= \begin{bmatrix}\mathbf{M}_{1,:}\\ \mathbf{M}_{2,:}\\ \mathbf{M}_{3,:}\end{bmatrix} && \text{(Theorem 0.1)}\\ &= \begin{bmatrix}1 & 2\\ 3 & -1\\ 0 & 4\end{bmatrix}, && \text{(substitute the rows)}\end{aligned}$$

so the second column of $\mathbf{M}$ lists the second entry of each row:

$$\mathbf{M}_{:,2} = \begin{bmatrix}2\\ -1\\ 4\end{bmatrix}.$$

Theorem 0.2 (A matrix is a row of its columns)**.** *An $m \times n$ matrix $\mathbf{A}$ is the row vector whose entries are its columns:*

$$\mathbf{A} = \begin{bmatrix}\mathbf{A}_{:,1} & \cdots & \mathbf{A}_{:,n}\end{bmatrix}.$$

Exercise 0.6 (Placing columns side by side)**.** Theorem 0.2 says that a matrix is determined by its columns, so measurements recorded one quantity at a time can be assembled into a matrix by placing those lists side by side. Let $\mathbf{N}$ be the 2×3 matrix with

$$\mathbf{N}_{:,1} = \begin{bmatrix}2\\ -1\end{bmatrix}, \qquad \mathbf{N}_{:,2} = \begin{bmatrix}0\\ 3\end{bmatrix}, \qquad \mathbf{N}_{:,3} = \begin{bmatrix}4\\ 1\end{bmatrix}.$$

What is $\mathbf{N}_{2,:}$, the second row of $\mathbf{N}$?

Hint

Place the columns from left to right, as in Theorem 0.2, then read across the second row.

Solution

Solution 0.6. Placing the three columns from left to right gives

$$\begin{aligned}\mathbf{N} &= \begin{bmatrix}\mathbf{N}_{:,1} & \mathbf{N}_{:,2} & \mathbf{N}_{:,3}\end{bmatrix} && \text{(Theorem 0.2)}\\ &= \begin{bmatrix}2 & 0 & 4\\ -1 & 3 & 1\end{bmatrix}, && \text{(substitute the columns)}\end{aligned}$$

so the second row of $\mathbf{N}$ lists the second entry of each column:

$$\mathbf{N}_{2,:} = \begin{bmatrix}-1 & 3 & 1\end{bmatrix}.$$

Definition 0.5 (Tensor)**.** A **tensor** [2] of order $k \geq 1$ is an array of numbers with k indices,

$$\mathcal{A} = \left(a_{i_1 i_2 \cdots i_k}\right), \quad i_1 = 1, \ldots, n_1, \ \ldots, \ i_k = 1, \ldots, n_k,$$

and its dimensions are $n_1 \times n_2 \times \cdots \times n_k$.

Example 0.3 (Vectors, matrices and images as tensors)**.** A vector of length p is a tensor of order 1, and an $m \times n$ matrix is a tensor of order 2. A 28×28 grayscale image is a matrix: the entry a_{ij} is the brightness of the pixel in row i and column j. A 28×28 color image stores three numbers per pixel (red, green, blue), so it is an order-3 tensor with dimensions $28 \times 28 \times 3$, and a_{ijc} is the value of color channel c at the pixel in row i and column j. It has $28 \cdot 28 \cdot 3 = 2352$ entries.

Exercise 0.7 (Counting the entries of a batch of photos)**.** Software for neural networks often stores a batch of color photos as one tensor, with one index for the photo, two for the pixel's position, and one for the color channel, and the software stores one number per entry, so the count of entries determines how much memory the batch needs. Let $\mathcal{B}$ be the order-4 tensor for a batch of 10 color photos of 32×32 pixels, with $n_1 = 10$, $n_2 = 32$, $n_3 = 32$ and $n_4 = 3$.
How many entries does $\mathcal{B}$ have?

Hint

Each index i_r in Definition 0.5 takes n_r values, and each choice of all four indices picks one entry, as in Example 0.3.

Solution

Solution 0.7. An entry of $\mathcal{B}$ is picked by choosing i_1 from 10 values, i_2 and i_3 from 32 values each, and i_4 from 3 values, so the number of entries is

$$\begin{aligned} n_1 \cdot n_2 \cdot n_3 \cdot n_4 &= 10 \cdot 32 \cdot 32 \cdot 3 && \text{(substitute the index ranges)} \\ &= 30720. && \text{(multiply)} \end{aligned}$$

Matrix transpose

Definition 0.1 (Matrix transpose)**.** The **transpose** [3] of an $m \times n$ matrix $\mathbf{A}$ is the $n \times m$ matrix $\mathbf{A}^\top$ obtained by swapping the rows and columns of $\mathbf{A}$:

$$(\mathbf{A}^\top)_{ij} = a_{ji}$$

Example 0.1 (Transposing a 2×3 matrix)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$,

$$\mathbf{A}^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix},$$

a 3×2 matrix. For instance, entry $(1, 2)$ of the transpose is $a_{21} = 4$.

Exercise 0.1 (Transposing a 3×2 matrix)**.** The transpose turns the rows of a matrix into columns, just as the transpose of a vector turns a column into a row. Let

$$\mathbf{B} = \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 4 & 5 \end{bmatrix}.$$

Compute $\mathbf{B}^\top$.

Hint

Entry (i, j) of the transpose is entry (j, i) of the original matrix, as in Example 0.1.

Solution

Solution 0.1. $\mathbf{B}$ is 3×2, so its transpose is 2×3. The ith row of $\mathbf{B}^\top$ is the ith column of $\mathbf{B}$:

$$\mathbf{B}^\top = \begin{bmatrix} 1 & 0 & 4 \\ -2 & 3 & 5 \end{bmatrix}.$$

For instance, entry $(2,3)$ of $\mathbf{B}^\top$ is $b_{32} = 5$.

Matrix addition

Definition 0.1 (Zero matrix)**.** The $m \times n$ **zero matrix** [4] $\mathbf{0}_{m\times n}$ (or $\mathbf{0}$ when dimensions are clear from context) has all entries equal to zero:

$$\mathbf{0}_{m\times n} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

Example 0.1 (Zero matrices of two shapes)**.**

$$\mathbf{0}_{2\times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \mathbf{0}_{3\times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Both have all entries 0, but one is 2×3 and the other 3×2, so they are different matrices.

Exercise 0.1 (Whether a matrix of zeros is a given zero matrix)**.** A zero matrix carries its shape in its subscript, because, as Example 0.1 shows, zero matrices of different shapes are different matrices. Let

$$\mathbf{Z} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Is $\mathbf{Z} = \mathbf{0}_{2\times 3}$?

Hint

Compare the shapes of the two matrices, using Definition 0.2.

Solution

Solution 0.1. No. $\mathbf{Z}$ has three rows and two columns, so it is 3×2. But $\mathbf{0}_{2\times 3}$ is 2×3. Two matrices of different shapes are not equal, even when every entry of both is 0. Instead, $\mathbf{Z}$ is the zero matrix $\mathbf{0}_{3\times 2}$.

Definition 0.2 (Matrix addition)**.** Two matrices $\mathbf{A}$ and $\mathbf{B}$ of the same dimensions $m\times n$ can be added element-wise; their **matrix sum** [5] is:

$$(\mathbf{A}+\mathbf{B})_{ij} = a_{ij} + b_{ij}$$

Example 0.2 (Adding matrices, and a sum that is not defined)**.**

$$\begin{aligned}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} + \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} &= \begin{bmatrix}1+0 & 2+1\\ 3+1 & 4+0\end{bmatrix} && \text{(add entry by entry)}\\ &= \begin{bmatrix}1 & 3\\ 4 & 4\end{bmatrix}. && \text{(add)}\end{aligned}$$

A 2×2 matrix and a 2×3 matrix cannot be added: they do not have the same dimensions, so entry $(1,3)$ of the sum would have no first term.

Exercise 0.2 (Adding two 2×2 matrices)**.** A matrix sum combines two tables of the same shape entry by entry, such as two months of counts recorded in the same rows and columns. Let

$$\mathbf{A} = \begin{bmatrix}1 & -1\\ 2 & 0\end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix}3 & 2\\ -2 & 5\end{bmatrix}.$$

Compute $\mathbf{A}+\mathbf{B}$.

Hint

Add the entries in the same position, as in Example 0.2.

Solution

Solution 0.2. Both matrices are 2×2, so the matrix sum is defined:

$$\begin{aligned}\mathbf{A}+\mathbf{B} &= \begin{bmatrix} 1+3 & -1+2 \\ 2+(-2) & 0+5 \end{bmatrix} && \text{(add entry by entry)} \\ &= \begin{bmatrix} 4 & 1 \\ 0 & 5 \end{bmatrix}. && \text{(add)}\end{aligned}$$

Theorem 0.1 (Matrix addition is commutative)**.** *For $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$, matrix addition is commutative:*

$$\mathbf{A}+\mathbf{B}=\mathbf{B}+\mathbf{A}$$

Exercise 0.3 (Proving that matrix addition is commutative)**.** Theorem 0.1 lets two matrices be added in either order, just as two numbers can be (sums are symmetric). Prove Theorem 0.1.

Hint

Two $m \times n$ matrices are equal when their (i, j) entries agree for every i and j. Express the (i, j) entry of each side with Definition 0.2.

Solution

Solution 0.3. Both sides are $m \times n$ matrices, so they are equal when their (i, j) entries agree for every i and j. For each i and j,

$$\begin{aligned}(\mathbf{A}+\mathbf{B})_{ij} &= a_{ij}+b_{ij} && \text{(Definition 0.2)} \\ &= b_{ij}+a_{ij} && \text{(sums of numbers are symmetric)} \\ &= (\mathbf{B}+\mathbf{A})_{ij}, && \text{(Definition 0.2)}\end{aligned}$$

where the middle step uses the symmetry of sums of numbers. So $\mathbf{A}+\mathbf{B}=\mathbf{B}+\mathbf{A}$.

Theorem 0.2 (Matrix addition is associative)**.** *For $m \times n$ matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, matrix addition is associative:*

$$(\mathbf{A}+\mathbf{B})+\mathbf{C}=\mathbf{A}+(\mathbf{B}+\mathbf{C})$$

Exercise 0.4 (Proving that matrix addition is associative)**.** Theorem 0.2 lets three matrices be added without parentheses, just as three numbers can be (sums are associative). Prove Theorem 0.2.

Hint

Two $m \times n$ matrices are equal when their (i, j) entries agree for every i and j. Apply Definition 0.2 twice on each side.

Solution

Solution 0.4. Both sides are $m \times n$ matrices, so they are equal when their (i, j) entries agree for every i and j. For each i and j,

$$\begin{aligned}
((\mathbf{A}+\mathbf{B})+\mathbf{C})_{ij} &= (\mathbf{A}+\mathbf{B})_{ij} + c_{ij} && \text{(Definition 0.2)} \\
&= (a_{ij}+b_{ij}) + c_{ij} && \text{(Definition 0.2)} \\
&= a_{ij} + (b_{ij}+c_{ij}) && \text{(sums of numbers are associative)} \\
&= a_{ij} + (\mathbf{B}+\mathbf{C})_{ij} && \text{(Definition 0.2)} \\
&= (\mathbf{A}+(\mathbf{B}+\mathbf{C}))_{ij}, && \text{(Definition 0.2)}
\end{aligned}$$

where the middle step uses the associativity of sums of numbers. So $(\mathbf{A}+\mathbf{B})+\mathbf{C} = \mathbf{A}+(\mathbf{B}+\mathbf{C})$.

Theorem 0.3 (Zero matrix is the additive identity)**.** *For an $m \times n$ matrix $\mathbf{A}$:*

$$\mathbf{A} + \mathbf{0}_{m\times n} = \mathbf{A}$$

Exercise 0.5 (Proving that the zero matrix is the additive identity)**.** Theorem 0.3 says that adding the zero matrix leaves a matrix unchanged, just as adding the number 0 leaves a number unchanged.
Prove Theorem 0.3.

Hint

Express the (i, j) entry of $\mathbf{A} + \mathbf{0}_{m\times n}$ with Definition 0.2, and use that every entry of $\mathbf{0}_{m\times n}$ is 0.

Solution

Solution 0.5. Both sides are $m \times n$ matrices, so they are equal when their (i, j) entries agree for every i and j. For each i and j,

$$\begin{aligned}
(\mathbf{A}+\mathbf{0}_{m\times n})_{ij} &= a_{ij} + (\mathbf{0}_{m\times n})_{ij} && \text{(Definition 0.2)} \\
&= a_{ij} + 0 && \text{(Definition 0.1)} \\
&= a_{ij}. && \text{(adding 0 to a number leaves it unchanged)}
\end{aligned}$$

So $\mathbf{A} + \mathbf{0}_{m\times n} = \mathbf{A}$.

Theorem 0.4 (Additive inverse)**.** *For any $m \times n$ matrix $\mathbf{A}$, the $m \times n$ matrix $-\mathbf{A}$ (defined by $(-\mathbf{A})_{ij} = -a_{ij}$) satisfies:*

$$\mathbf{A} + (-\mathbf{A}) = \mathbf{0}_{m \times n}$$

Exercise 0.6 (Proving that a matrix plus its negation is zero)**.** Theorem 0.4 says that adding $-\mathbf{A}$ to a matrix $\mathbf{A}$ gives the zero matrix, just as adding $-a$ to a number a gives 0. Here $-\mathbf{A}$, the negation of $\mathbf{A}$, negates every entry of $\mathbf{A}$.
Prove Theorem 0.4.

Hint

Express the (i, j) entry of $\mathbf{A} + (-\mathbf{A})$ with Definition 0.2, and use $(-\mathbf{A})_{ij} = -a_{ij}$ from Theorem 0.4.

Solution

Solution 0.6. Both sides are $m \times n$ matrices, so they are equal when their (i, j) entries agree for every i and j. For each i and j,

$$\begin{aligned}
(\mathbf{A} + (-\mathbf{A}))_{ij} &= a_{ij} + (-\mathbf{A})_{ij} && \text{(Definition 0.2)} \\
&= a_{ij} + (-a_{ij}) && \text{(definition of } -\mathbf{A} \text{ in Theorem 0.4)} \\
&= 0 && \text{(a number plus its negative is 0)} \\
&= (\mathbf{0}_{m \times n})_{ij}. && \text{(Definition 0.1)}
\end{aligned}$$

So $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}_{m \times n}$.

Scalar multiplication

Definition 0.1 (Scalar multiplication)**.** The **scalar multiple** [6] of a matrix $\mathbf{A}$ by a scalar c is:

$$(c\mathbf{A})_{ij} = c \cdot a_{ij}$$

Example 0.1 (A scalar multiple)**.**

$$3\begin{bmatrix}1 & -2\\ 0 & 4\end{bmatrix} = \begin{bmatrix}3\cdot 1 & 3\cdot(-2)\\ 3\cdot 0 & 3\cdot 4\end{bmatrix} = \begin{bmatrix}3 & -6\\ 0 & 12\end{bmatrix}.$$

Exercise 0.1 (Multiplying a matrix by a negative factor)**.** A scalar multiple rescales every entry of a matrix by the same factor, as when a table of distances is converted from meters to centimeters. A negative factor flips the sign of every entry, and a factor between -1 and 1 shrinks every entry toward 0. Let

$$\mathbf{C} = \begin{bmatrix}4 & -2\\ 0 & 6\end{bmatrix}.$$

Compute $-\frac{1}{2}\mathbf{C}$.

Hint

Multiply each entry by $-\frac{1}{2}$, as in Example 0.1.

Solution

Solution 0.1.

$$-\frac{1}{2}\mathbf{C} = \begin{bmatrix}-\frac{1}{2}\cdot 4 & -\frac{1}{2}\cdot(-2)\\ -\frac{1}{2}\cdot 0 & -\frac{1}{2}\cdot 6\end{bmatrix} \quad \text{(Definition 0.1)}$$

$$= \begin{bmatrix}-2 & 1\\ 0 & -3\end{bmatrix}. \quad \text{(multiply)}$$

Matrix multiplication

Definition 0.1 (Matrix multiplication)**.** The **product** [7] of an $m \times k$ matrix $\mathbf{A}$ and a $k \times n$ matrix $\mathbf{B}$ is the $m \times n$ matrix $\mathbf{C} = \mathbf{AB}$ with entries:

$$c_{ij} = \sum_{s=1}^{k} a_{is}\, b_{sj}$$

Remark (Matrix multiplication is not commutative)

Remark 0.1 (Matrix multiplication is not commutative). When **AB** and **BA** are both defined, they are usually different. For example, with

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

$$\mathbf{AB} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

$$\mathbf{BA} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix},$$

so $\mathbf{AB} \neq \mathbf{BA}$.

Example 0.1 (Dot product as matrix multiplication)**.** The dot product of two column vectors $\underset{\sim}{x}$ and $\underset{\sim}{\beta}$ can be written as a matrix product of the row vector $\underset{\sim}{x}^\top$ with the column vector $\underset{\sim}{\beta}$:

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{\beta} &= \underset{\sim}{x}^\top \underset{\sim}{\beta} \\ &= [x_1,\ x_2,\ \ldots,\ x_p] \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} \\ &= x_1\beta_1 + x_2\beta_2 + \cdots + x_p\beta_p \end{aligned}$$

Exercise 0.1 (Computing the product of a 2×3 and a 3×2 matrix)**.** The product of two matrices is defined only when the first has as many columns as the second has rows. Each entry of the product combines one row of the first factor with one column of the second. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 4 \end{bmatrix}.$$

Find **AB**.

Hint

A is 2×3, and **B** is 3×2. By Definition 0.1, **AB** is therefore 2×2. Its entry c_{ij} is $a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j}$.

Solution

Solution 0.1. **A** has three columns, and **B** has three rows. So the product is defined and is 2×2. Its entries are

$$\begin{aligned}\mathbf{AB} &= \begin{bmatrix} 1\cdot 2+0\cdot 0+2\cdot 1 & 1\cdot 1+0\cdot(-1)+2\cdot 4 \\ (-1)\cdot 2+3\cdot 0+1\cdot 1 & (-1)\cdot 1+3\cdot(-1)+1\cdot 4 \end{bmatrix} && \text{(Definition 0.1)} \\ &= \begin{bmatrix} 4 & 9 \\ -1 & 0 \end{bmatrix}. && \text{(multiply and add)}\end{aligned}$$

Theorem 0.1 (Matrix multiplication is associative)**.** *For an $m \times k$ matrix* **A***, a $k \times l$ matrix* **B***, and an $l \times n$ matrix* **C***:*

$$\underbrace{(\mathbf{AB})\mathbf{C}}_{m\times n} = \underbrace{\mathbf{A}(\mathbf{BC})}_{m\times n}$$

Proof

Proof. Entry (i, j) of each side, from Definition 0.1:

$$\begin{aligned}[(\mathbf{AB})\mathbf{C}]_{ij} &= \sum_{t=1}^{l} (\mathbf{AB})_{it}\, c_{tj} && \text{(definition of } (\mathbf{AB})\mathbf{C}) \\ &= \sum_{t=1}^{l} \left(\sum_{s=1}^{k} a_{is}\, b_{st} \right) c_{tj} && \text{(definition of } \mathbf{AB}) \\ &= \sum_{s=1}^{k} a_{is} \left(\sum_{t=1}^{l} b_{st}\, c_{tj} \right) && \text{(distribute and swap the finite sums)} \\ &= \sum_{s=1}^{k} a_{is}\, (\mathbf{BC})_{sj} && \text{(definition of } \mathbf{BC}) \\ &= [\mathbf{A}(\mathbf{BC})]_{ij} && \text{(definition of } \mathbf{A}(\mathbf{BC}))\end{aligned}$$

□

Exercise 0.2 (Grouping a product of three matrices)**.** By Theorem 0.1, a product of three matrices can be grouped either way, so $\mathbf{AB}\underset{\sim}{x}$ can be computed as $\mathbf{A}(\mathbf{B}\underset{\sim}{x})$, without ever forming **AB**. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \qquad \underset{\sim}{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

where $\underset{\sim}{x}$ is a 2×1 matrix.

Find $\mathbf{AB}\underset{\sim}{x}$ by computing $\mathbf{B}\underset{\sim}{x}$ first.

Hint

By Definition 0.1, $\mathbf{B}\underset{\sim}{x}$ is a 2×1 matrix; multiply $\mathbf{A}$ by it.

Solution

Solution 0.2. First,

$$\begin{aligned} \mathbf{B}\underset{\sim}{x} &= \begin{bmatrix} 2 \cdot 1 + 0 \cdot 2 \\ 1 \cdot 1 + 1 \cdot 2 \end{bmatrix} && \text{(Definition 0.1)} \\ &= \begin{bmatrix} 2 \\ 3 \end{bmatrix}, && \text{(multiply and add)} \end{aligned}$$

so

$$\begin{aligned} \mathbf{AB}\underset{\sim}{x} &= \mathbf{A}(\mathbf{B}\underset{\sim}{x}) && \text{(Theorem 0.1)} \\ &= \begin{bmatrix} 1 \cdot 2 + 1 \cdot 3 \\ 0 \cdot 2 + 1 \cdot 3 \end{bmatrix} && \text{(Definition 0.1)} \\ &= \begin{bmatrix} 5 \\ 3 \end{bmatrix}. && \text{(multiply and add)} \end{aligned}$$

Grouping the other way gives the same answer: $\mathbf{AB} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$, and $(\mathbf{AB})\underset{\sim}{x} = \begin{bmatrix} 3+2 \\ 1+2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$.

Theorem 0.2 (Matrix multiplication is distributive over addition)**.** *Matrix multiplication distributes over addition from the left and from the right. For an $m \times k$ matrix $\mathbf{A}$ and $k \times n$ matrices $\mathbf{B}$ and $\mathbf{C}$:*

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

For $m \times k$ matrices $\mathbf{A}$ and $\mathbf{B}$ and a $k \times n$ matrix $\mathbf{C}$:

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$$

Exercise 0.3 (Proving that matrix multiplication distributes from the left)**.** Theorem 0.2 lets a product with a matrix sum be expanded term by term, just as multiplication of numbers distributes over addition. Let $\mathbf{A}$ be an $m \times k$ matrix, and let $\mathbf{B}$ and $\mathbf{C}$ be $k \times n$ matrices.

Prove that $\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC}$.

Hint

Express the (i,j) entry of the left side with Definition 0.1 and Definition 0.2, then split the result with the rule for a sum of sums.

Solution

Solution 0.3. Both sides are $m \times n$ matrices, so they are equal when their (i,j) entries agree for every i and j. For each i and j,

$$\begin{aligned}
[\mathbf{A}(\mathbf{B}+\mathbf{C})]_{ij} &= \sum_{s=1}^{k} a_{is}(\mathbf{B}+\mathbf{C})_{sj} && \text{(Definition 0.1)} \\
&= \sum_{s=1}^{k} a_{is}(b_{sj}+c_{sj}) && \text{(Definition 0.2)} \\
&= \sum_{s=1}^{k} (a_{is}b_{sj} + a_{is}c_{sj}) && \text{(multiplication of numbers distributes over addition)} \\
&= \sum_{s=1}^{k} a_{is}b_{sj} + \sum_{s=1}^{k} a_{is}c_{sj} && \text{(a sum of sums splits)} \\
&= (\mathbf{AB})_{ij} + (\mathbf{AC})_{ij} && \text{(Definition 0.1)} \\
&= (\mathbf{AB}+\mathbf{AC})_{ij}. && \text{(Definition 0.2)}
\end{aligned}$$

The third step uses that multiplication of numbers distributes over addition, and the fourth uses the rule for a sum of sums. So $\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC}$.

Theorem 0.3 (Product of two sums of matrices)**.** *For $m \times k$ matrices $\mathbf{A}$ and $\mathbf{B}$ and $k \times n$ matrices $\mathbf{C}$ and $\mathbf{D}$,*

$$(\mathbf{A}+\mathbf{B})(\mathbf{C}+\mathbf{D}) = \mathbf{AC}+\mathbf{AD}+\mathbf{BC}+\mathbf{BD}.$$

Each product keeps its factors in the order they appear on the left, because matrix multiplication is not commutative. So $\mathbf{AD}$ and $\mathbf{BC}$ are not collected into $2\mathbf{AD}$.

Proof

Proof.

$$\begin{aligned}(\mathbf{A}+\mathbf{B})(\mathbf{C}+\mathbf{D}) &= (\mathbf{A}+\mathbf{B})\mathbf{C}+(\mathbf{A}+\mathbf{B})\mathbf{D} && \text{(Theorem 0.2, from the left)}\\ &= \mathbf{AC}+\mathbf{BC}+\mathbf{AD}+\mathbf{BD} && \text{(Theorem 0.2, from the right, twice)}\\ &= \mathbf{AC}+\mathbf{AD}+\mathbf{BC}+\mathbf{BD} && \text{(matrix addition is commutative)}\end{aligned}$$

□

Example 0.2 (Why the middle terms do not combine)**.** Take $\mathbf{A}=\mathbf{C}=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ and $\mathbf{B}=\mathbf{D}=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$. Then $\mathbf{A}+\mathbf{B}=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$, so

$$\begin{aligned}(\mathbf{A}+\mathbf{B})^2 &= \begin{bmatrix}5 & 4\\ 4 & 5\end{bmatrix},\\ \mathbf{A}^2+\mathbf{AB}+\mathbf{BA}+\mathbf{B}^2 &= \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}+\begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}+\begin{bmatrix}1 & 1\\ 1 & 2\end{bmatrix}+\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}\\ &= \begin{bmatrix}5 & 4\\ 4 & 5\end{bmatrix}.\end{aligned}$$

Here $\mathbf{AB}\neq\mathbf{BA}$, so the middle terms do not combine into $2\mathbf{AB}$. The squared norm of a sum of vectors, $\|\underset{\sim}{x}+\underset{\sim}{y}\|^2$, is a case where they do combine (Theorem 2 in Inner Products and Orthogonality), because $\underset{\sim}{x}\cdot\underset{\sim}{y}=\underset{\sim}{y}\cdot\underset{\sim}{x}$. Matrices that commute, such as $\mathbf{A}$ and $\mathbf{A}^2$, combine the same way.

Exercise 0.4 (Comparing $(\mathbf{A}+\mathbf{B})(\mathbf{A}+\mathbf{B})$ with the rule for numbers)**.** For numbers, $(a+b)^2=a^2+2ab+b^2$. Theorem 0.3 expands a product of two matrix sums into four terms instead, because $\mathbf{AB}$ and $\mathbf{BA}$ can differ. Let

$$\mathbf{A}=\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix},\qquad \mathbf{B}=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}.$$

Is $(\mathbf{A}+\mathbf{B})(\mathbf{A}+\mathbf{B})$ equal to $\mathbf{AA}+2\mathbf{AB}+\mathbf{BB}$?

Hint

Compute $\mathbf{A}+\mathbf{B}$ and its product with itself, then compute $\mathbf{AA}$, $\mathbf{AB}$ and $\mathbf{BB}$ with Definition 0.1.

Solution

Solution 0.4. No. On the left, $\mathbf{A}+\mathbf{B}=\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}$, so

$$(\mathbf{A}+\mathbf{B})(\mathbf{A}+\mathbf{B}) = \begin{bmatrix}1\cdot 1+1\cdot 0 & 1\cdot 1+1\cdot 0\\ 0\cdot 1+0\cdot 0 & 0\cdot 1+0\cdot 0\end{bmatrix} \quad \text{(Definition 0.1)}$$
$$= \begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}. \quad \text{(multiply and add)}$$

On the right, the same rule gives $\mathbf{AA}=\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$, $\mathbf{AB}=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$ and $\mathbf{BB}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$, so

$$\mathbf{AA}+2\mathbf{AB}+\mathbf{BB}=\begin{bmatrix}1 & 2\\ 0 & 0\end{bmatrix}.$$

The entries in position $(1,2)$ differ, $1\neq 2$, so the two matrices are not equal. The rule for numbers fails because $\mathbf{BA}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$ differs from $\mathbf{AB}$: Theorem 0.3 gives $\mathbf{AA}+\mathbf{AB}+\mathbf{BA}+\mathbf{BB}=\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}$, which does match the left side.

Exercise 0.5 (Which of these products are defined?)**.** Let $\mathbf{A}$ be 4×3, let $\mathbf{B}$ be 3×2, and let $\underset{\sim}{x}\in\mathbb{R}^3$. For each expression, say whether it is defined, and if so give the shape of the result:

1. $\mathbf{AB}$
2. $\mathbf{BA}$
3. $\mathbf{A}\underset{\sim}{x}$
4. $\underset{\sim}{x}^{\top}\underset{\sim}{x}$
5. $\underset{\sim}{x}\underset{\sim}{x}^{\top}$

Solution

Solution 0.5. A vector in $\mathbb{R}^3$ is a 3×1 matrix, so the same rule settles every case.

1. $\mathbf{AB}$: $(4\times 3)(3\times 2)$. The inner pair is 3 and 3, so it is defined, and the result is 4×2.
2. $\mathbf{BA}$: $(3\times 2)(4\times 3)$. The inner pair is 2 and 4, which do not match, so it is **not defined**. Matrix multiplication is not commutative, and this is the blunt form of that: swapping the order can leave an expression that means nothing.
3. $\mathbf{A}\underset{\sim}{x}$: $(4\times 3)(3\times 1)$, defined, result 4×1 — a vector in $\mathbb{R}^4$.
4. $\underset{\sim}{x}^{\top}\underset{\sim}{x}$: $(1\times 3)(3\times 1)$, defined, result 1×1 — a single number. The product is the inner product of Equation 1 in Vectors.

5. $\underset{\sim}{x}\underset{\sim}{x}^\top$: $(3 \times 1)(1 \times 3)$, defined, result 3×3 — a matrix.

The last two use the same two vectors and differ only in order, and they return objects of different kinds. Reading a transpose as decoration rather than as a shape change is the commonest way to lose track of an expression.

Square and identity matrices

Definition 0.1 (Square matrix)**.** A matrix is **square** [8] if it has the same number of rows as columns.

Example 0.1 (Square and not square)**.** $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is square (2 rows, 2 columns); $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is not (2 rows, 3 columns).

Exercise 0.1 (Whether a data matrix times its transpose is square)**.** Many statistical methods start from an $n \times p$ data matrix $\mathbf{X}$, with one row per observation, and form the product $\mathbf{X}^\top\mathbf{X}$ of its transpose with itself. Whether that product is square matters, because some operations, such as taking the inverse of a matrix, apply only to square matrices.
Is $\mathbf{X}^\top\mathbf{X}$ square?

> **Hint**
>
> Find the shape of $\mathbf{X}^\top$ from Definition 0.1, then the shape of the product from Definition 0.1.

Solution

Solution 0.1. Yes. The transpose $\mathbf{X}^\top$ is $p \times n$, so $\mathbf{X}^\top$ has n columns. $\mathbf{X}$ has n rows, so the product $\mathbf{X}^\top\mathbf{X}$ is defined and is $p \times p$. It has p rows and p columns, so it is square for every n and p.

Definition 0.2 (Main diagonal)**.** The **main diagonal** [9] of an $m \times n$ matrix $\mathbf{A}$ is the list of its entries $a_{11}, a_{22}, \ldots, a_{kk}$, where k is the smaller of m and n.

Definition 0.3 (Diagonal entry)**.** The entries on the main diagonal (Definition 0.2) of a matrix $\mathbf{A}$ are the **diagonal entries** of $\mathbf{A}$.

Definition 0.4 (Off-diagonal entry)**.** In an $m \times n$ matrix $\mathbf{A}$ with main diagonal (Definition 0.2), every entry a_{ij} with $i \neq j$ is an **off-diagonal entry**.

Example 0.2 (Diagonal entries of a square and a non-square matrix)**.**

- For $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, the diagonal entries are $a_{11} = 1$ and $a_{22} = 4$, and the off-diagonal entries are $a_{12} = 2$ and $a_{21} = 3$.
- For $\mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $m = 2$ and $n = 3$, so $k = 2$, and the diagonal entries are $b_{11} = 1$ and $b_{22} = 5$; the other four entries, 2, 3, 4 and 6, are off-diagonal.

Exercise 0.2 (The main diagonal of a tall matrix)**.** The main diagonal of a matrix that is not square stops when it runs out of rows or columns, whichever comes first. A tall matrix, with more rows than columns, runs out of columns first. Let

$$\mathbf{E} = \begin{bmatrix} 7 & -2 \\ 1 & 3 \\ 0 & 5 \\ 4 & -6 \end{bmatrix}.$$

Find the main diagonal of $\mathbf{E}$.

Hint

Find m, n and k in Definition 0.2.

Solution

Solution 0.2. Write e_{ij} for the entry of $\mathbf{E}$ in row i and column j. $\mathbf{E}$ is 4×2, so $m = 4$, $n = 2$ and $k = 2$. The main diagonal of $\mathbf{E}$ is therefore the list e_{11}, e_{22}, that is, $7, 3$. The list stops at e_{22} because $\mathbf{E}$ has no third column. It therefore has no entry e_{33}.

Exercise 0.3 (Adding the diagonal entries of a square matrix)**.** The sum of the diagonal entries of a square matrix summarizes that matrix in one number, called its trace. Let

$$\mathbf{F} = \begin{bmatrix} 2 & 5 & -1 \\ 0 & -3 & 8 \\ 4 & 1 & 6 \end{bmatrix}.$$

Find the sum of the diagonal entries of $\mathbf{F}$.

Hint

Find the main diagonal of $\mathbf{F}$ first.

Solution

Solution 0.3. Write f_{ij} for the entry of $\mathbf{F}$ in row i and column j. $\mathbf{F}$ is 3×3, so its diagonal entries are $f_{11} = 2$, $f_{22} = -3$ and $f_{33} = 6$, and adding them gives

$$f_{11} + f_{22} + f_{33} = 2 + (-3) + 6 = 5.$$

Exercise 0.4 (Counting the off-diagonal entries of a square matrix)**.** In a table of pairwise comparisons among several items, the off-diagonal entries are the comparisons between two different items, so counting them counts the ordered pairs of different items. Let $\mathbf{A}$ be a 5×5 matrix, so $\mathbf{A}$ is square.
How many off-diagonal entries does $\mathbf{A}$ have?

Hint

Count all the entries, then subtract the diagonal entries.

Solution

Solution 0.4. $\mathbf{A}$ has $5 \cdot 5 = 25$ entries. Its diagonal entries are the 5 entries a_{ii}, $i = 1, \ldots, 5$, and every other entry has $i \neq j$, so $\mathbf{A}$ has $25 - 5 = 20$ off-diagonal entries.

Definition 0.5 (Identity matrix)**.** The $p \times p$ **identity matrix** [10] $\mathbf{I}_p$ (or $\mathbf{I}$ when the size is clear from context) has ones on the main diagonal and zeros elsewhere:

$$(\mathbf{I}_p)_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \qquad \mathbf{I}_p = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Example 0.3 (The 3×3 identity)**.**

$$\mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

so, for instance, $(\mathbf{I}_3)_{22} = 1$ and $(\mathbf{I}_3)_{23} = 0$.

Exercise 0.5 (Whether a matrix with the identity pattern is an identity matrix)**.** An identity matrix has ones on its main diagonal and zeros elsewhere. The matrix below has the same pattern of ones and zeros. Let

$$\mathbf{J} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Is $\mathbf{J}$ an identity matrix?

Hint

Compare the shape of $\mathbf{J}$ with the shape of $\mathbf{I}_p$ in Definition 0.5.

Solution

Solution 0.5. No. Every identity matrix $\mathbf{I}_p$ is $p \times p$, so it is square. $\mathbf{J}$ is 2×3, so it is not square. Therefore $\mathbf{J}$ is not an identity matrix, even though its main diagonal holds ones and its other entries are zeros.

Remark (The identity matrix and the Kronecker delta)

Remark 0.1 (The identity matrix and the Kronecker delta). The entries of the identity matrix are given by the Kronecker delta: $(\mathbf{I}_p)_{ij} = \delta_{ij}$.

Theorem 0.1 (Identity matrix is a multiplicative identity)**.** *For any $m \times p$ matrix $\mathbf{A}$:*

$$\mathbf{A}\,\mathbf{I}_p = \mathbf{A}$$

$$\mathbf{I}_m\,\mathbf{A} = \mathbf{A}$$

Exercise 0.6 (Multiplying by an identity matrix)**.** Theorem 0.1 says that an identity matrix of the right size leaves a matrix unchanged under matrix multiplication, just as

multiplying a number by 1 leaves it unchanged. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 3 & 4 \end{bmatrix}.$$

Compute $\mathbf{I}_2\mathbf{A}$ entry by entry, from the definition of matrix multiplication.

Hint

Entry (i, j) of $\mathbf{I}_2\mathbf{A}$ is the dot product of the ith row of $\mathbf{I}_2$ with the jth column of $\mathbf{A}$, as in Example 0.1.

Solution

Solution 0.6. $\mathbf{I}_2$ is 2×2 and $\mathbf{A}$ is 2×3, so the product is defined and is 2×3. The first row of $\mathbf{I}_2$ is $\begin{bmatrix} 1 & 0 \end{bmatrix}$ and the second is $\begin{bmatrix} 0 & 1 \end{bmatrix}$, so by Definition 0.1

$$\begin{aligned} \mathbf{I}_2\mathbf{A} &= \begin{bmatrix} 1\cdot 2 + 0\cdot 5 & 1\cdot 0 + 0\cdot 3 & 1\cdot(-1) + 0\cdot 4 \\ 0\cdot 2 + 1\cdot 5 & 0\cdot 0 + 1\cdot 3 & 0\cdot(-1) + 1\cdot 4 \end{bmatrix} && \text{(Definition 0.1)} \\ &= \begin{bmatrix} 2 & 0 & -1 \\ 5 & 3 & 4 \end{bmatrix} && \text{(multiply and add)} \\ &= \mathbf{A}. \end{aligned}$$

Each row of $\mathbf{I}_2$ has a single 1, so entry (i, j) of the product picks out entry (i, j) of $\mathbf{A}$, as Theorem 0.1 says.

Matrix-vector multiplication

Definition 0.1 (Matrix-vector multiplication)**.** The **matrix-vector product** of an $m \times p$ matrix $\mathbf{A}$ and a $p \times 1$ column vector $\underset{\sim}{x}$ is the $m \times 1$ column vector $\mathbf{A}\underset{\sim}{x}$ with entries:

$$(\mathbf{A}\underset{\sim}{x})_i = \sum_{j=1}^{p} a_{ij}\, x_j$$

Remark (Each entry is a dot product)

Remark 0.1 (Each entry is a dot product). Matrix-vector multiplication extends the dot product (Definition 8 in Vectors): entry i of $\mathbf{A}\underset{\sim}{x}$ is the dot product of row i of $\mathbf{A}$ with $\underset{\sim}{x}$. For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot (-1) \\ 3 \cdot 1 + 4 \cdot (-1) \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

Exercise 0.1 (A matrix-vector product with more columns than rows)**.** A matrix-vector product $\mathbf{A}\underset{\sim}{x}$ computes one weighted sum of the entries of $\underset{\sim}{x}$ for each row of $\mathbf{A}$, so one multiplication computes several weighted sums at once. The matrix $\mathbf{A}$ below has more columns than rows, so $\mathbf{A}\underset{\sim}{x}$ is shorter than $\underset{\sim}{x}$. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 3 & 4 \end{bmatrix}, \qquad \underset{\sim}{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Compute $\mathbf{A}\underset{\sim}{x}$.

Hint

Entry i of $\mathbf{A}\underset{\sim}{x}$ is the dot product of the ith row of $\mathbf{A}$ with $\underset{\sim}{x}$, as in Remark 0.1.

Solution

Solution 0.1. $\mathbf{A}$ is 2×3 and $\underset{\sim}{x}$ is 3×1, so the matrix-vector product is defined and is 2×1:

$$\begin{aligned} \mathbf{A}\underset{\sim}{x} &= \begin{bmatrix} 2 \cdot 1 + 0 \cdot 2 + (-1) \cdot 3 \\ 5 \cdot 1 + 3 \cdot 2 + 4 \cdot 3 \end{bmatrix} && \text{(Definition 0.1)} \\ &= \begin{bmatrix} 2 + 0 - 3 \\ 5 + 6 + 12 \end{bmatrix} && \text{(multiply)} \\ &= \begin{bmatrix} -1 \\ 23 \end{bmatrix}. && \text{(add)} \end{aligned}$$

Definition 0.2 (Linear system, system of linear equations)**.** A **linear system** [11] (or **system of linear equations**) with m equations in n unknowns is an equation $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ in which the $m \times n$ matrix $\mathbf{A}$ and the vector $\underset{\sim}{b} \in \mathbb{R}^m$ are given. A **solution** of the system is a vector $\underset{\sim}{x} \in \mathbb{R}^n$ for which $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ holds.

Example 0.1 (Two equations in two unknowns)**.** By Definition 0.1, row i of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ is the equation $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$. So the two equations $x_1 + 2x_2 = 5$ and $3x_1 + 4x_2 = 6$

form the linear system $\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \underset{\sim}{x} = \begin{bmatrix}5\\ 6\end{bmatrix}$. The vector $\underset{\sim}{x} = (-4, 4.5)$ is a solution:

$$\begin{aligned}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}-4\\ 4.5\end{bmatrix} &= \begin{bmatrix}1\cdot(-4) + 2\cdot 4.5\\ 3\cdot(-4) + 4\cdot 4.5\end{bmatrix} && \text{(Definition 0.1)}\\ &= \begin{bmatrix}-4+9\\ -12+18\end{bmatrix} && \text{(multiply)}\\ &= \begin{bmatrix}5\\ 6\end{bmatrix} && \text{(add)}\end{aligned}$$

The vector $(1, 2)$ is not a solution: its first equation holds, since $1 + 2 \cdot 2 = 5$, but its second gives $3 \cdot 1 + 4 \cdot 2 = 11 \neq 6$.

Exercise 0.2 (Checking whether a vector solves a linear system)**.** Checking whether a vector solves a linear system takes only one matrix-vector product, which is much less work than solving the system. Let

$$\mathbf{A} = \begin{bmatrix}3 & 1\\ 1 & -2\end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix}5\\ 4\end{bmatrix}, \qquad \underset{\sim}{x} = \begin{bmatrix}2\\ -1\end{bmatrix}.$$

Is $\underset{\sim}{x}$ a solution of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$?

Hint

Compute $\mathbf{A}\underset{\sim}{x}$ with Definition 0.1. Then compare it with $\underset{\sim}{b}$, as in Example 0.1.

Solution

Solution 0.2. Yes. The matrix-vector product is

$$\begin{aligned}\mathbf{A}\underset{\sim}{x} &= \begin{bmatrix}3\cdot 2 + 1\cdot(-1)\\ 1\cdot 2 + (-2)\cdot(-1)\end{bmatrix} && \text{(Definition 0.1)}\\ &= \begin{bmatrix}5\\ 4\end{bmatrix}, && \text{(multiply and add)}\end{aligned}$$

which matches $\underset{\sim}{b}$ entry by entry, so $\underset{\sim}{x}$ is a solution of the system.

Linear and affine maps

Definition 0.1 (Additive function)**.** A function $f : \mathbb{R}^p \to \mathbb{R}^m$ is **additive** [12] if it preserves vector addition: for all $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$,

$$f(\underset{\sim}{x} + \underset{\sim}{y}) = f(\underset{\sim}{x}) + f(\underset{\sim}{y})$$

Example 0.1 (An additive function, and one that is not)**.**

- $f(x) = 2x$ on $\mathbb{R}$ is additive. For $x, y \in \mathbb{R}$:

$$\begin{aligned} f(x+y) &= 2(x+y) \\ &= 2x + 2y \\ &= f(x) + f(y) \end{aligned}$$

- $g(x) = x + 1$ on $\mathbb{R}$ is not additive, since $g(0+0) = 1$ but $g(0) + g(0) = 2$.

Exercise 0.1 (Testing whether $x_1 x_2$ is additive)**.** An additive function can be evaluated at $\underset{\sim}{x} + \underset{\sim}{y}$ by evaluating it at $\underset{\sim}{x}$ and at $\underset{\sim}{y}$ separately, so additivity is worth checking before splitting a computation that way. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be $f(\underset{\sim}{x}) = x_1 x_2$.
Is f additive?

Hint

Try $\underset{\sim}{x} = (1, 0)$ and $\underset{\sim}{y} = (0, 1)$ in Definition 0.1.

Solution

Solution 0.1. No. For $\underset{\sim}{x} = (1, 0)$ and $\underset{\sim}{y} = (0, 1)$, $\underset{\sim}{x} + \underset{\sim}{y} = (1, 1)$, so

$$f(\underset{\sim}{x} + \underset{\sim}{y}) = 1 \cdot 1 = 1, \qquad f(\underset{\sim}{x}) + f(\underset{\sim}{y}) = 1 \cdot 0 + 0 \cdot 1 = 0.$$

These differ, so f does not satisfy Definition 0.1 for this pair of inputs, and f is not additive.

Definition 0.2 (Homogeneous function)**.** A function $f : \mathbb{R}^p \to \mathbb{R}^m$ is **homogeneous** [12] if it preserves scalar multiplication: for all $\underset{\sim}{x} \in \mathbb{R}^p$ and $c \in \mathbb{R}$,

$$f(c\underset{\sim}{x}) = cf(\underset{\sim}{x})$$

Example 0.2 (Homogeneous functions, and one that is not)**.**

- $f(x) = 2x$ on $\mathbb{R}$ is homogeneous. For $x, c \in \mathbb{R}$:

$$\begin{aligned} f(cx) &= 2cx \\ &= c(2x) \\ &= cf(x) \end{aligned}$$

- $h(x) = x^2$ on $\mathbb{R}$ is not homogeneous, since $h(2 \cdot 1) = 4$ but $2\,h(1) = 2$.
- $k(\underset{\sim}{x}) = (x_1^3 + x_2^3)^{1/3}$ on $\mathbb{R}^2$ is homogeneous but not additive. For $\underset{\sim}{x} \in \mathbb{R}^2$ and $c \in \mathbb{R}$:

$$\begin{aligned} k(c\underset{\sim}{x}) &= (c^3x_1^3 + c^3x_2^3)^{1/3} \\ &= (c^3(x_1^3 + x_2^3))^{1/3} \\ &= c(x_1^3 + x_2^3)^{1/3} \\ &= c\,k(\underset{\sim}{x}) \end{aligned}$$

  but $k((1,0) + (0,1)) = k((1,1)) = 2^{1/3}$, while $k((1,0)) + k((0,1)) = 2$.

Exercise 0.2 (Testing whether the absolute value is homogeneous)**.** A homogeneous function turns a rescaled input into the same rescaling of its output, including a rescaling by a negative number. Let $f : \mathbb{R} \to \mathbb{R}$ be the absolute value, $f(x) = |x|$.
Is f homogeneous?

Hint

Try a negative c in Definition 0.2.

Solution

Solution 0.2. No. Take $c = -1$ and $x = 1$. Then

$$f(cx) = |-1| = 1, \qquad cf(x) = -1 \cdot |1| = -1.$$

These differ, so f is not homogeneous. It does satisfy $f(cx) = cf(x)$ for every $c \geq 0$, since then $|cx| = c|x|$.

Definition 0.3 (Linear map, linear transformation, linear function)**.** A function $f : \mathbb{R}^p \to \mathbb{R}^m$ is a **linear map** [12] (also called a **linear transformation** or **linear function**) if it is both additive and homogeneous.

Example 0.3 (A linear map, and two functions that are not)**.**

- $f(\underset{\sim}{x}) = (2x_1,\ x_1 + x_2)$ on $\mathbb{R}^2$ is linear. For $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^2$ and $c \in \mathbb{R}$:

$$\begin{aligned} f(\underset{\sim}{x} + \underset{\sim}{y}) &= (2(x_1 + y_1),\ (x_1 + y_1) + (x_2 + y_2)) \\ &= f(\underset{\sim}{x}) + f(\underset{\sim}{y}) \end{aligned}$$

  and

$$\begin{aligned} f(c\underset{\sim}{x}) &= (2cx_1,\ cx_1 + cx_2) \\ &= cf(\underset{\sim}{x}). \end{aligned}$$

- $g(x) = x+1$ on $\mathbb{R}$ is not linear: additivity fails, since $g(0+0) = 1$ but $g(0)+g(0) = 2$.
- $h(x) = x^2$ on $\mathbb{R}$ is not linear: $h(2 \cdot 1) = 4$ but $2\,h(1) = 2$, so $h(cx) \neq c\,h(x)$ for $c = 2$, $x = 1$.

Exercise 0.3 (Testing whether a difference of scaled measurements is a linear map)**.** A score that scales two measurements and takes their difference, such as $3x_1 - x_2$, is a common building block of a model. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be $f(\underset{\sim}{x}) = 3x_1 - x_2$.
Is f a linear map?

Hint

Check the two conditions in Definition 0.3 one at a time, for any $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^2$ and $c \in \mathbb{R}$.

Solution

Solution 0.3. Yes. For any $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^2$, f is additive:

$$\begin{aligned} f(\underset{\sim}{x} + \underset{\sim}{y}) &= 3(x_1 + y_1) - (x_2 + y_2) && \text{(Definition 7 in Vectors)} \\ &= (3x_1 - x_2) + (3y_1 - y_2) && \text{(rearranging the terms)} \\ &= f(\underset{\sim}{x}) + f(\underset{\sim}{y}). \end{aligned}$$

For any $\underset{\sim}{x} \in \mathbb{R}^2$ and $c \in \mathbb{R}$, f is homogeneous:

$$\begin{aligned} f(c\underset{\sim}{x}) &= 3cx_1 - cx_2 && \text{(Definition 0.1)} \\ &= c(3x_1 - x_2) && \text{(factoring out } c\text{)} \\ &= cf(\underset{\sim}{x}). \end{aligned}$$

Since f is both, it is a linear map.

Definition 0.4 (Linear operator)**.** A **linear operator** on $\mathbb{R}^p$ is a linear map (Definition 0.3) from $\mathbb{R}^p$ to itself, that is, a linear map $f : \mathbb{R}^p \to \mathbb{R}^p$ whose input and output have the same length. Some authors use "operator" more loosely, for any linear map.

Example 0.4 (A linear operator, and a linear map that is not one)**.** The map $f(\underset{\sim}{x}) = (2x_1,\ x_1 + x_2)$ from Example 0.3 sends $\mathbb{R}^2$ to $\mathbb{R}^2$, so it is a linear operator on $\mathbb{R}^2$. The linear map $g(\underset{\sim}{x}) = x_1 + x_2$ sends $\mathbb{R}^2$ to $\mathbb{R}$, so it is linear but not an operator: its outputs have length 1, not 2.

Exercise 0.4 (Testing whether a projection is a linear operator)**.** Dropping the last coordinate of a point in $\mathbb{R}^3$ projects it onto a plane, as a shadow does. Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be $f(\underset{\sim}{x}) = (x_1, x_2)$. This f is a linear map.
Is f a linear operator on $\mathbb{R}^3$?

Hint

Write f as $f : \mathbb{R}^p \to \mathbb{R}^m$, and compare p with m as in Definition 0.4.

Solution

Solution 0.4. No. A linear operator on $\mathbb{R}^3$ sends $\mathbb{R}^3$ to $\mathbb{R}^3$. The outputs of f lie in $\mathbb{R}^2$, not $\mathbb{R}^3$, so f is a linear map but not a linear operator.

Theorem 0.1 (A linear map sends zero to zero)**.** *If $f : \mathbb{R}^p \to \mathbb{R}^m$ is a linear map (Definition 0.3), then $f(\underset{\sim}{0}) = \underset{\sim}{0}$.*

Proof

Proof. On the left, $\underset{\sim}{0}$ is the zero vector of length p; on the right, it is the zero vector of length m.

$$\begin{aligned} f(\underset{\sim}{0}) &= f(0 \cdot \underset{\sim}{0}) && \text{(every entry of } 0 \cdot \underset{\sim}{0} \text{ is } 0 \cdot 0 = 0\text{; Definition 0.1)} \\ &= 0 \cdot f(\underset{\sim}{0}) && \text{(homogeneity, with } c = 0 \text{ and } \underset{\sim}{x} = \underset{\sim}{0}\text{)} \\ &= \underset{\sim}{0} && \text{(every entry of } 0 \cdot f(\underset{\sim}{0}) \text{ is 0; Definition 0.1)} \end{aligned}$$

□

Example 0.5 (Ruling out linearity at zero)**.** In Example 0.3, $g(x) = x + 1$ has $g(0) = 1 \neq 0$, so by Theorem 0.1 it is not linear, without checking additivity or homogeneity. Sending $\underset{\sim}{0}$ to $\underset{\sim}{0}$ is necessary but not sufficient: $h(x) = x^2$ in the same example has $h(0) = 0$

and is still not linear.

Exercise 0.5 (Ruling out a linear map from its value at zero)**.** Theorem 0.1 gives a quick test: a function that does not send $\underset{\sim}{0}$ to $\underset{\sim}{0}$ cannot be a linear map. Let $g : \mathbb{R}^2 \to \mathbb{R}^2$ be

$$g(\underset{\sim}{x}) = (x_1 + x_2,\ x_2 - 1).$$

Is g a linear map?

Hint

Compute $g(\underset{\sim}{0})$.

Solution

Solution 0.5. No.

$$g(\underset{\sim}{0}) = (0 + 0,\ 0 - 1) = (0, -1) \neq \underset{\sim}{0}.$$

Every linear map sends $\underset{\sim}{0}$ to $\underset{\sim}{0}$ by Theorem 0.1, so g is not one.

Theorem 0.2 (Multiplying by a matrix is a linear map)**.** *For any $m \times p$ matrix $\mathbf{A}$, the function $f : \mathbb{R}^p \to \mathbb{R}^m$ defined by $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ is a linear map (Definition 0.3).*

Proof

Proof. Both conditions are checked one entry at a time. For each $i \in \{1, \ldots, m\}$, additivity:

$$\begin{aligned}
(\mathbf{A}(\underset{\sim}{x} + \underset{\sim}{y}))_i &= \sum_{j=1}^{p} a_{ij}(\underset{\sim}{x} + \underset{\sim}{y})_j && \text{(Definition 0.1)} \\
&= \sum_{j=1}^{p} a_{ij}(x_j + y_j) && \text{(Definition 7 in Vectors)} \\
&= \sum_{j=1}^{p} (a_{ij}x_j + a_{ij}y_j) && \text{(multiplication distributes over addition)} \\
&= \sum_{j=1}^{p} a_{ij}x_j + \sum_{j=1}^{p} a_{ij}y_j && \text{(a sum of sums splits)} \\
&= (\mathbf{A}\underset{\sim}{x})_i + (\mathbf{A}\underset{\sim}{y})_i && \text{(Definition 0.1)} \\
&= (\mathbf{A}\underset{\sim}{x} + \mathbf{A}\underset{\sim}{y})_i && \text{(Definition 7 in Vectors)}
\end{aligned}$$

and homogeneity:

$$\begin{aligned}
(\mathbf{A}(c\underset{\sim}{x}))_i &= \sum_{j=1}^{p} a_{ij}(c\underset{\sim}{x})_j && \text{(Definition 0.1)} \\
&= \sum_{j=1}^{p} a_{ij}\, c\, x_j && \text{(Definition 0.1)} \\
&= \sum_{j=1}^{p} c\, a_{ij} x_j && \text{(products are symmetric)} \\
&= c \sum_{j=1}^{p} a_{ij} x_j && \text{(a constant factor comes out of a sum)} \\
&= c\,(\mathbf{A}\underset{\sim}{x})_i && \text{(Definition 0.1)} \\
&= (c\,\mathbf{A}\underset{\sim}{x})_i && \text{(Definition 0.1)}
\end{aligned}$$

The four algebra rules cited here are on the algebra page: distributivity, a sum of sums, symmetry of products, and a constant factor. □

Exercise 0.6 (Recognizing a linear map as multiplication by a matrix)**.** Theorem 0.2 saves checking additivity and homogeneity by hand: a function that multiplies its input by a fixed matrix is a linear map. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be

$$f(\underset{\sim}{x}) = (x_1 + 2x_2,\ 3x_1 - x_2).$$

Is f a linear map?

Hint

Find a 2×2 matrix $\mathbf{A}$ with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$, using Definition 0.1.

Solution

Solution 0.6. Yes. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}.$$

By Definition 0.1,

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} 1 \cdot x_1 + 2 \cdot x_2 \\ 3 \cdot x_1 + (-1) \cdot x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - x_2 \end{bmatrix} = f(\underset{\sim}{x}).$$

So f multiplies its input by $\mathbf{A}$, and f is a linear map by Theorem 0.2.

Theorem 0.3 (A linear map preserves linear combinations)**.** *If $f : \mathbb{R}^p \to \mathbb{R}^m$ is a linear map (Definition 0.3), then for any vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathbb{R}^p$ and any coefficients $c_1, \ldots, c_k \in \mathbb{R}$:*

$$f\left(\sum_{j=1}^{k} c_j \underset{\sim}{v}_j\right) = \sum_{j=1}^{k} c_j f(\underset{\sim}{v}_j)$$

Proof

Proof. By induction on k. For $k = 1$, the claim $f(c_1 \underset{\sim}{v}_1) = c_1 f(\underset{\sim}{v}_1)$ is homogeneity. Suppose the claim holds for k vectors. Then for $k+1$:

$$\begin{aligned}
f\left(\sum_{j=1}^{k+1} c_j \underset{\sim}{v}_j\right) &= f\left(\sum_{j=1}^{k} c_j \underset{\sim}{v}_j + c_{k+1} \underset{\sim}{v}_{k+1}\right) && \text{(split off the last term)} \\
&= f\left(\sum_{j=1}^{k} c_j \underset{\sim}{v}_j\right) + f(c_{k+1} \underset{\sim}{v}_{k+1}) && \text{(additivity)} \\
&= \sum_{j=1}^{k} c_j f(\underset{\sim}{v}_j) + f(c_{k+1} \underset{\sim}{v}_{k+1}) && \text{(induction hypothesis)} \\
&= \sum_{j=1}^{k} c_j f(\underset{\sim}{v}_j) + c_{k+1} f(\underset{\sim}{v}_{k+1}) && \text{(homogeneity)} \\
&= \sum_{j=1}^{k+1} c_j f(\underset{\sim}{v}_j) && \text{(join the last term back on)}
\end{aligned}$$

□

Exercise 0.7 (Evaluating a linear map on a linear combination)**.** A linear map is determined on a linear combination of vectors by its values on those vectors alone. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map with

$$f(\underset{\sim}{v}_1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad f(\underset{\sim}{v}_2) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Find $f(3\underset{\sim}{v}_1 - 2\underset{\sim}{v}_2)$.

Hint

Apply Theorem 0.3 with $c_1 = 3$ and $c_2 = -2$.

Solution

Solution 0.7.

$$\begin{aligned} f(3\underset{\sim}{v}_1 - 2\underset{\sim}{v}_2) &= 3f(\underset{\sim}{v}_1) - 2f(\underset{\sim}{v}_2) && \text{(Theorem 0.3)} \\ &= 3\begin{bmatrix}1\\2\end{bmatrix} - 2\begin{bmatrix}0\\-1\end{bmatrix} && \text{(the given values)} \\ &= \begin{bmatrix}3\\6\end{bmatrix} - \begin{bmatrix}0\\-2\end{bmatrix} && \text{(Definition 0.1)} \\ &= \begin{bmatrix}3\\8\end{bmatrix} && \text{(subtracting entry by entry)} \end{aligned}$$

Theorem 0.4 (Every linear map is multiplication by a matrix)**.** *If $f : \mathbb{R}^p \to \mathbb{R}^m$ is a linear map (Definition 0.3), then there is exactly one $m \times p$ matrix $\mathbf{A}$ such that*

$$f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} \quad \text{for all } \underset{\sim}{x} \in \mathbb{R}^p$$

Column j of $\mathbf{A}$ is $f(\underset{\sim}{e}_j)$, the value of f at the jth indicator vector (Definition 18 in Vectors); that is, $a_{ij} = \big(f(\underset{\sim}{e}_j)\big)_i$.

Proof

Proof. **Existence.** Let $\mathbf{A}$ be the $m \times p$ matrix with $a_{ij} = \big(f(\underset{\sim}{e}_j)\big)_i$. For any $\underset{\sim}{x} \in \mathbb{R}^p$,

$$\begin{aligned} f(\underset{\sim}{x}) &= f\left(\sum_{j=1}^{p} x_j \underset{\sim}{e}_j\right) && \text{(Theorem 4 in Vectors)} \\ &= \sum_{j=1}^{p} x_j f(\underset{\sim}{e}_j) && \text{(Theorem 0.3)} \end{aligned}$$

and entry i of that sum is

$$\begin{aligned} \left(\sum_{j=1}^{p} x_j f(\underset{\sim}{e}_j)\right)_i &= \sum_{j=1}^{p} x_j \big(f(\underset{\sim}{e}_j)\big)_i && \text{(Definition 9 in Vectors, entry by entry)} \\ &= \sum_{j=1}^{p} x_j a_{ij} && \text{(definition of } \mathbf{A}) \\ &= \sum_{j=1}^{p} a_{ij} x_j && \text{(products are symmetric)} \\ &= (\mathbf{A}\underset{\sim}{x})_i && \text{(Definition 0.1)} \end{aligned}$$

Uniqueness. Suppose $\mathbf{B}$ is any $m \times p$ matrix with $f(\underset{\sim}{x}) = \mathbf{B}\underset{\sim}{x}$ for all $\underset{\sim}{x}$. Taking $\underset{\sim}{x} = \underset{\sim}{e}_j$, for each i:

$$\begin{aligned}
\big(f(\underset{\sim}{e}_j)\big)_i &= (\mathbf{B}\underset{\sim}{e}_j)_i && \text{(assumption on } \mathbf{B}\text{)} \\
&= \sum_{k=1}^{p} b_{ik}(\underset{\sim}{e}_j)_k && \text{(Definition 0.1)} \\
&= b_{ij} \cdot 1 + \sum_{k \neq j} b_{ik} \cdot 0 && \text{(Definition 18 in Vectors, splitting off the } k = j \text{ term)} \\
&= b_{ij} + 0 && \text{(times 1 changes nothing; times 0 gives 0)} \\
&= b_{ij} && \text{(adding 0 changes nothing)}
\end{aligned}$$

So

$$\begin{aligned}
b_{ij} &= \big(f(\underset{\sim}{e}_j)\big)_i \\
&= a_{ij}
\end{aligned}$$

for every i and j, and $\mathbf{B} = \mathbf{A}$. □

Definition 0.5 (Matrix of a linear map)**.** The **matrix of** a linear map $f : \mathbb{R}^p \to \mathbb{R}^m$ is the unique $m \times p$ matrix $\mathbf{A}$ with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ for all $\underset{\sim}{x}$ (Theorem 0.4). Its column j is $f(\underset{\sim}{e}_j)$.

Example 0.6 (Finding the matrix of a linear map)**.** For $f(\underset{\sim}{x}) = (2x_1,\ x_1 + x_2)$ from Example 0.3, the values of f at the indicator vectors are

$$f(\underset{\sim}{e}_1) = f\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}2 \cdot 1\\1 + 0\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix}, \qquad f(\underset{\sim}{e}_2) = f\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}2 \cdot 0\\0 + 1\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$$

and these are the columns of the matrix of f (Definition 0.5): $\mathbf{A} = \begin{bmatrix}2 & 0\\1 & 1\end{bmatrix}$.

Exercise 0.8 (Evaluating a linear map from its values on the indicator vectors)**.** By Theorem 0.4, the values of a linear map at the indicator vectors determine it everywhere. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map with

$$f(\underset{\sim}{e}_1) = \begin{bmatrix}1\\3\end{bmatrix}, \qquad f(\underset{\sim}{e}_2) = \begin{bmatrix}-2\\0\end{bmatrix}.$$

Find $f\left(\begin{bmatrix}4\\1\end{bmatrix}\right)$.

Hint

Build the matrix of f from the two given values, then multiply.

Solution

Solution 0.8. By Theorem 0.4, $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$, where column j of $\mathbf{A}$ is $f(\underset{\sim}{e}_j)$:

$$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}.$$

So

$$f\left(\begin{bmatrix} 4 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \cdot 4 + (-2) \cdot 1 \\ 3 \cdot 4 + 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \end{bmatrix}.$$

Exercise 0.9 (Finding the matrix of a linear map into $\mathbb{R}^3$)**.** The matrix of a linear map turns the map into a table of numbers that software can store and multiply. Let $f : \mathbb{R}^2 \to \mathbb{R}^3$ be the linear map

$$f(\underset{\sim}{x}) = (x_2,\ x_1,\ x_1 - x_2).$$

Find the matrix of f.

Hint

Column j of the matrix is $f(\underset{\sim}{e}_j)$.

Solution

Solution 0.9. The values of f at the indicator vectors are

$$f(\underset{\sim}{e}_1) = f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad f(\underset{\sim}{e}_2) = f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

These values are the columns of the matrix of f:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & -1 \end{bmatrix}.$$

It is 3×2, since f sends $\mathbb{R}^2$ to $\mathbb{R}^3$.

Corollary 0.1 (Linear functions from vectors to numbers)**.** *A function $f : \mathbb{R}^p \to \mathbb{R}$ is linear (Definition 0.3) if and only if there is a vector $\underset{\sim}{w} \in \mathbb{R}^p$ such that*

$$f(\underset{\sim}{x}) = \underset{\sim}{w} \cdot \underset{\sim}{x} \quad \text{for all } \underset{\sim}{x} \in \mathbb{R}^p$$

and then $w_j = f(\underset{\sim}{e}_j)$ for each j.

Proof

Proof. **If f is linear.** Theorem 0.4 with $m = 1$ gives a $1 \times p$ matrix $\mathbf{A}$ with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ and $a_{1j} = f(\underset{\sim}{e}_j)$. Let $\underset{\sim}{w}$ be the vector with $w_j = a_{1j}$, so that $\mathbf{A} = \underset{\sim}{w}^\top$. Then

$$\begin{aligned} f(\underset{\sim}{x}) &= \mathbf{A}\underset{\sim}{x} && \text{(Theorem 0.4)} \\ &= \underset{\sim}{w} \cdot \underset{\sim}{x} && (\mathbf{A} = \underset{\sim}{w}^\top) \end{aligned}$$

and

$$\begin{aligned} w_j &= a_{1j} \\ &= f(\underset{\sim}{e}_j). \end{aligned}$$

If $f(\underset{\sim}{x}) = \underset{\sim}{w}^\top \underset{\sim}{x}$. Here $\underset{\sim}{w}^\top$ is a $1 \times p$ matrix, so f is linear by Theorem 0.2. □

Example 0.7 (A linear function from $\mathbb{R}^2$ to $\mathbb{R}$)**.** $f(\underset{\sim}{x}) = 3x_1 + 5x_2$ is $\underset{\sim}{w} \cdot \underset{\sim}{x}$ with $\underset{\sim}{w} = (3, 5)^\top$, so it is linear by Corollary 0.1. The entries of $\underset{\sim}{w}$ are the values at the indicator vectors:

$$\begin{aligned} f(\underset{\sim}{e}_1) &= 3 \cdot 1 + 5 \cdot 0 \\ &= 3 \end{aligned}$$

and

$$\begin{aligned} f(\underset{\sim}{e}_2) &= 3 \cdot 0 + 5 \cdot 1 \\ &= 5. \end{aligned}$$

$g(\underset{\sim}{x}) = 3x_1 + 5x_2 + 1$ is not linear, because $g(\underset{\sim}{0}) = 1 \neq 0$ (Theorem 0.1).

Exercise 0.10 (Finding the weight vector of a linear score)**.** A linear score of p measurements is a dot product of the measurements with a vector of weights, by Corollary 0.1. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a linear map with

$$f(\underset{\sim}{e}_1) = 2, \qquad f(\underset{\sim}{e}_2) = 0, \qquad f(\underset{\sim}{e}_3) = -1.$$

Find the vector $\underset{\sim}{w}$ with $f(\underset{\sim}{x}) = \underset{\sim}{w} \cdot \underset{\sim}{x}$ for every $\underset{\sim}{x} \in \mathbb{R}^3$.

Hint

Corollary 0.1 gives each entry w_j of $\underset{\sim}{w}$.

Solution

Solution 0.10. By Corollary 0.1, $w_j = f(\underset{\sim}{e}_j)$ for each j, so

$$\underset{\sim}{w} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}.$$

Then $f(\underset{\sim}{x}) = 2x_1 - x_3$ for every $\underset{\sim}{x} \in \mathbb{R}^3$.

Videos

Two chapters of the *Essence of linear algebra* series by 3Blue1Brown show the results above geometrically:

- Linear transformations and matrices (Chapter 3) shows why a linear map is fixed by where it sends the indicator vectors, which become the columns of its matrix (Definition 0.5).
- Dot products and duality (Chapter 9) shows why every linear map from $\mathbb{R}^p$ to $\mathbb{R}$ is a dot product with a fixed vector (Corollary 0.1).

Definition 0.6 (Translation)**.** For a fixed vector $\underset{\sim}{b} \in \mathbb{R}^m$, the **translation** [13] by $\underset{\sim}{b}$ is the function $t_{\underset{\sim}{b}} : \mathbb{R}^m \to \mathbb{R}^m$ defined by

$$t_{\underset{\sim}{b}}(\underset{\sim}{y}) = \underset{\sim}{y} + \underset{\sim}{b}$$

Example 0.8 (Translating two points)**.** The translation by $\underset{\sim}{b} = (1, -2)^\top$ sends

$$t_{\underset{\sim}{b}}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0+1 \\ 0+(-2) \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \qquad t_{\underset{\sim}{b}}\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 3+1 \\ 4+(-2) \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Both points move one unit right and two units down.

Remark (A translation moves every point by the same vector)

Remark 0.1 (A translation moves every point by the same vector). A translation by $\underset{\sim}{b}$ moves every point by the same vector $\underset{\sim}{b}$.

Exercise 0.11 (Finding the translation that moves one point to another)**.** A translation moves every point by the same vector, so where it sends one point fixes the whole translation. A translation $t_{\underset{\sim}{b}}$ of $\mathbb{R}^2$ sends $(2,3)^\top$ to $(5,1)^\top$.
Find $\underset{\sim}{b}$.

Hint

Solve $(2,3)^\top + \underset{\sim}{b} = (5,1)^\top$ entry by entry.

Solution

Solution 0.11. By Definition 0.6, $t_{\underset{\sim}{b}}\left((2,3)^\top\right) = (2,3)^\top + \underset{\sim}{b}$. Matching this vector with $(5,1)^\top$ entry by entry gives

$$\underset{\sim}{b} = \begin{bmatrix} 5-2 \\ 1-3 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

Definition 0.7.

Affine map, affine transformation, affine function
A function $f : \mathbb{R}^p \to \mathbb{R}^m$ is an **affine map** (also called an **affine transformation** or **affine function**) if it is a matrix product followed by a translation (Definition 0.6) (a composition, with the matrix product as the inner function): there are an $m \times p$ matrix $\mathbf{A}$ and a vector $\underset{\sim}{b} \in \mathbb{R}^m$ such that

$$\begin{aligned} f(\underset{\sim}{x}) &= t_{\underset{\sim}{b}}(\mathbf{A}\underset{\sim}{x}) \\ &= \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b} \quad \text{for all } \underset{\sim}{x} \in \mathbb{R}^p \end{aligned}$$

Exercise 0.12 (Testing whether a rescale-and-shift is an affine map)**.** Rescaling a measurement and then shifting it, as when converting units with a different zero point, is the typical affine map. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be

$$f(\underset{\sim}{x}) = (2x_1 + 1,\ x_2 - 3).$$

Is f an affine map?

Hint

Look for a 2×2 matrix $\mathbf{A}$ and a vector $\underset{\sim}{b}$ with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$.

Solution

Solution 0.12. Yes. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}.$$

By Definition 0.1, $\mathbf{A}\underset{\sim}{x} = (2x_1, x_2)^\top$, so

$$\mathbf{A}\underset{\sim}{x} + \underset{\sim}{b} = \begin{bmatrix} 2x_1 + 1 \\ x_2 - 3 \end{bmatrix} = f(\underset{\sim}{x})$$

for every $\underset{\sim}{x} \in \mathbb{R}^2$. So f is an affine map.

Definition 0.8 (Offset of an affine map, intercept, bias)**.** In Definition 0.7, the vector $\underset{\sim}{b}$ is called the **offset** of f (statistics also calls it the **intercept**, and machine learning the **bias**).

Remark (Real-valued affine maps)

Remark 0.2 (Real-valued affine maps). For a real-valued function (Definition 4 in Vectors, $m = 1$), $\mathbf{A}$ is a single row, written $\underset{\sim}{w}^\top$, and $\underset{\sim}{b}$ is a single number b, so an affine function is a linear function $\underset{\sim}{w} \cdot \underset{\sim}{x}$ (Corollary 0.1) plus an offset:

$$f(\underset{\sim}{x}) = \underset{\sim}{w} \cdot \underset{\sim}{x} + b, \qquad \underset{\sim}{w} \in \mathbb{R}^p, \quad b \in \mathbb{R} \tag{1}$$

Exercise 0.13 (Reading off the offset of an affine map)**.** In a fitted model, the offset is the part of the output that does not depend on the input. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the affine map

$$f(\underset{\sim}{x}) = (x_1 - x_2 + 4,\ 3x_2 - 1).$$

Find the offset of f.

Hint

Write $f(\underset{\sim}{x})$ as $\mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ as in Definition 0.7.

Solution

Solution 0.13.

$$f(\underset{\sim}{x}) = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix} \underset{\sim}{x} + \begin{bmatrix} 4 \\ -1 \end{bmatrix},$$

so the offset of f is $\underset{\sim}{b} = (4, -1)^\top$. Each entry of $\underset{\sim}{b}$ is the term with no x_j in the matching entry of $f(\underset{\sim}{x})$.

Definition 0.9 (Augmented vector)**.** The **augmented vector** of $\underset{\sim}{x} \in \mathbb{R}^p$ is the vector $\underset{\sim}{x}^{\text{aug}} \in \mathbb{R}^{p+1}$ formed by placing a 1 above the entries of $\underset{\sim}{x}$:

$$\begin{aligned} \underset{\sim}{x}^{\text{aug}} &= \begin{bmatrix} 1 \\ \underset{\sim}{x} \end{bmatrix} \\ &= (1, x_1, \dots, x_p)^\top \end{aligned}$$

Example 0.9 (An affine function as a dot product with an augmented vector)**.** For $\underset{\sim}{x} = (2, 5)^\top$, the augmented vector is $\underset{\sim}{x}^{\text{aug}} = (1, 2, 5)^\top$. With $\underset{\sim}{w} = (3, -1)^\top$ and $b = 4$, the affine function $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ (Equation 1) equals the dot product of $(b, w_1, w_2)^\top = (4, 3, -1)^\top$ with $\underset{\sim}{x}^{\text{aug}}$:

$$\begin{aligned} \underset{\sim}{w} \cdot \underset{\sim}{x} + b &= 3 \cdot 2 + (-1) \cdot 5 + 4 \\ &= 5, \\ (4, 3, -1)^\top \cdot \underset{\sim}{x}^{\text{aug}} &= 4 \cdot 1 + 3 \cdot 2 + (-1) \cdot 5 \\ &= 5 \end{aligned}$$

Exercise 0.14 (Absorbing an offset into a dot product with the augmented vector)**.** The augmented vector lets software treat an affine score as a dot product, with the offset as one more weight. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be

$$f(\underset{\sim}{x}) = 2x_1 - x_2 + 5.$$

Find the vector $\underset{\sim}{v} \in \mathbb{R}^3$ with $f(\underset{\sim}{x}) = \underset{\sim}{v} \cdot \underset{\sim}{x}^{\text{aug}}$ for every $\underset{\sim}{x} \in \mathbb{R}^2$.

Hint

Follow Example 0.9. The first entry of $\underset{\sim}{x}^{\text{aug}}$ is 1.

Solution

Solution 0.14. Let $\underset{\sim}{v} = (5, 2, -1)^\top$. Since $\underset{\sim}{x}^{\text{aug}} = (1, x_1, x_2)^\top$ by Definition 0.9,

$$\underset{\sim}{v} \cdot \underset{\sim}{x}^{\text{aug}} = 5 \cdot 1 + 2x_1 + (-1)x_2 = f(\underset{\sim}{x}).$$

The first entry of $\underset{\sim}{v}$ is the offset 5, and the other two are the numbers multiplying x_1 and x_2.

Theorem 0.5 (An affine map sends zero to its offset)**.** *An affine map $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ (Definition 0.7) satisfies $f(\underset{\sim}{0}) = \underset{\sim}{b}$.*

Proof

Proof. First, $\mathbf{A}\underset{\sim}{0} = \underset{\sim}{0}$: for each i,

$$\begin{aligned}
(\mathbf{A}\underset{\sim}{0})_i &= \sum_{j=1}^{p} a_{ij} \cdot 0 && \text{(Definition 0.1)} \\
&= \sum_{j=1}^{p} 0 && \text{(any number times 0 is 0)} \\
&= 0 && \text{(a sum of zeros is 0)}
\end{aligned}$$

Then

$$\begin{aligned}
f(\underset{\sim}{0}) &= \mathbf{A}\underset{\sim}{0} + \underset{\sim}{b} && \text{(Definition 0.7)} \\
&= \underset{\sim}{0} + \underset{\sim}{b} && (\mathbf{A}\underset{\sim}{0} = \underset{\sim}{0}) \\
&= \underset{\sim}{b} && \text{(adding } \underset{\sim}{0} \text{ changes no entry; Definition 7 in Vectors)}
\end{aligned}$$

□

Exercise 0.15 (Recovering a column of an affine map's matrix from two values)**.** Theorem 0.5 says that the value of an affine map at $\underset{\sim}{0}$ is its offset, so subtracting that value isolates the matrix part. Let $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ be an affine map from $\mathbb{R}^2$ to $\mathbb{R}^2$ with

$$f(\underset{\sim}{0}) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

Find the first column of $\mathbf{A}$.

Hint

$\mathbf{A}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is the first column of $\mathbf{A}$.

Solution

Solution 0.15. By Theorem 0.5, $\underset{\sim}{b} = f(\underset{\sim}{0}) = (1,2)^\top$. So

$$\begin{aligned}\mathbf{A}\begin{bmatrix}1\\0\end{bmatrix} &= f\left(\begin{bmatrix}1\\0\end{bmatrix}\right) - \underset{\sim}{b} && \text{(Definition 0.7)}\\ &= \begin{bmatrix}4-1\\2-2\end{bmatrix} && \text{(the given values)}\\ &= \begin{bmatrix}3\\0\end{bmatrix}.\end{aligned}$$

By Definition 0.1, $\mathbf{A}(1,0)^\top$ is the first column of $\mathbf{A}$, which is therefore $(3,0)^\top$.

Theorem 0.6 (An affine map is linear exactly when its offset is zero)**.** *An affine map $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ (Definition 0.7) is linear (Definition 0.3) if and only if $\underset{\sim}{b} = \underset{\sim}{0}$.*

Proof

Proof. If $\underset{\sim}{b} = \underset{\sim}{0}$, then

$$\begin{aligned}f(\underset{\sim}{x}) &= \mathbf{A}\underset{\sim}{x} + \underset{\sim}{0}\\ &= \mathbf{A}\underset{\sim}{x},\end{aligned}$$

which is linear by Theorem 0.2. Conversely, if f is linear, then $f(\underset{\sim}{0}) = \underset{\sim}{0}$ by Theorem 0.1, and $f(\underset{\sim}{0}) = \underset{\sim}{b}$ by Theorem 0.5, so $\underset{\sim}{b} = \underset{\sim}{0}$. □

Example 0.10 (Two affine maps with the same matrix)**.** Let $\mathbf{A} = \begin{bmatrix}2 & 0\\1 & 1\end{bmatrix}$, the matrix of the linear map in Example 0.3. $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ has offset $\underset{\sim}{0}$, so it is linear. $g(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \begin{bmatrix}1\\0\end{bmatrix}$ has offset $\begin{bmatrix}1\\0\end{bmatrix} \neq \underset{\sim}{0}$, so by Theorem 0.6 it is not linear: indeed $g(\underset{\sim}{0}) = \begin{bmatrix}1\\0\end{bmatrix}$, not $\underset{\sim}{0}$.

Exercise 0.16 (Choosing a constant that makes an affine map linear)**.** By Theorem 0.6, an affine map is a linear map exactly when its offset is $\underset{\sim}{0}$. For $c \in \mathbb{R}$, let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be

$$f(\underset{\sim}{x}) = (x_1 - x_2 + c - 2,\ 3x_1 + c^2 - 4).$$

For which values of c is f linear?

Hint

Find the offset of f as a function of c.

Solution

Solution 0.16. Only for $c = 2$. The map is

$$f(\underset{\sim}{x}) = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix} \underset{\sim}{x} + \begin{bmatrix} c-2 \\ c^2-4 \end{bmatrix},$$

so its offset is $(c-2,\ c^2-4)^\top$. By Theorem 0.6, f is linear exactly when both entries are 0. The first entry is 0 only for $c = 2$, and then the second entry is $2^2 - 4 = 0$ too.

Corollary 0.2 (Every linear map is affine)**.** *Every linear map $f : \mathbb{R}^p \to \mathbb{R}^m$ (Definition 0.3) is an affine map (Definition 0.7) with offset $\underset{\sim}{0}$.*

Proof

Proof. If f is linear, Theorem 0.4 gives a matrix $\mathbf{A}$ with

$$\begin{aligned} f(\underset{\sim}{x}) &= \mathbf{A}\underset{\sim}{x} \\ &= \mathbf{A}\underset{\sim}{x} + \underset{\sim}{0}, \end{aligned}$$

which is Definition 0.7 with $\underset{\sim}{b} = \underset{\sim}{0}$. □

Exercise 0.17 (Testing whether a linear score is an affine map)**.** Some results hold only for affine maps, so it helps to check whether a given score qualifies. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be $f(\underset{\sim}{x}) = x_1 - 4x_2$, a linear map.
Is f an affine map?

Hint

Use Corollary 0.2.

Solution

Solution 0.17. Yes. By Corollary 0.2, every linear map is an affine map with offset $\underset{\sim}{0}$. Here

$$f(\underset{\sim}{x}) = \begin{bmatrix} 1 & -4 \end{bmatrix} \underset{\sim}{x} + 0,$$

with the 1×2 matrix $\begin{bmatrix} 1 & -4 \end{bmatrix}$ and offset 0.

Theorem 0.7 (An affine map determines its matrix and offset)**.** *If* $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ *and* $f(\underset{\sim}{x}) = \mathbf{A}'\underset{\sim}{x} + \underset{\sim}{b}'$ *for all* $\underset{\sim}{x} \in \mathbb{R}^p$ *(Definition 0.7), then* $\mathbf{A}' = \mathbf{A}$ *and* $\underset{\sim}{b}' = \underset{\sim}{b}$*.*

Proof

Proof. By Theorem 0.5, applied to each formula,

$$\begin{aligned} \underset{\sim}{b} &= f(\underset{\sim}{0}) \\ &= \underset{\sim}{b}'. \end{aligned}$$

Then

$$\begin{aligned} \mathbf{A}\underset{\sim}{x} &= f(\underset{\sim}{x}) - \underset{\sim}{b} \\ &= \mathbf{A}'\underset{\sim}{x} \end{aligned}$$

for all $\underset{\sim}{x}$, so $\mathbf{A}$ and $\mathbf{A}'$ are both matrices (Definition 0.5) of the linear map $\underset{\sim}{x} \mapsto f(\underset{\sim}{x}) - \underset{\sim}{b}$ (linear by Theorem 0.2), and the uniqueness in Theorem 0.4 gives $\mathbf{A}' = \mathbf{A}$. □

Exercise 0.18 (Matching the offsets of two formulas for one affine map)**.** Theorem 0.7 says that an affine map has only one matrix and one offset, so two ways of writing the same map must agree term by term. For $c \in \mathbb{R}$, suppose that

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \underset{\sim}{x} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \underset{\sim}{x} + \begin{bmatrix} c \\ 1 \end{bmatrix} \quad \text{for every } \underset{\sim}{x} \in \mathbb{R}^2.$$

Find c.

Hint

Compare the offsets of the two sides.

Solution

Solution 0.18. $c = 3$. Both sides describe the same affine map. By Theorem 0.7, the two sides have the same offset, so $(3, 1)^\top = (c, 1)^\top$ and $c = 3$. Setting $\underset{\sim}{x} = \underset{\sim}{0}$ gives the same answer directly.

Remark ("Linear" in this site, and in machine learning and statistics)

Remark 0.3 ("Linear" in this site, and in machine learning and statistics). This site uses "linear" only in the sense of Definition 0.3, so by Theorem 0.6 a function with a nonzero offset is affine but not linear. For functions of one variable, this is the same split as affine $ax + b$ versus linear ax on the algebra page. (The algebra page writes these as $mx + b$

and mx; here m already names the number of outputs.)
Machine learning and statistics often call any function of the form Equation 1 "linear", as in a *linear model* in statistics or a *linear layer* in a neural network, and call the offset b the bias (machine learning) or the intercept (statistics), as Definition 0.7 notes. A model with an intercept is affine in $\underset{\sim}{x}$ but linear in its coefficients $(b, \underset{\sim}{w})$, and that is the sense of "linear" in "linear model": it stays linear in the coefficients even when the inputs are transformed: $w_1 x + w_2 x^2 + b$ is not affine in x, but it is linear in (b, w_1, w_2). Separately, the affine function $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ is a linear function of the augmented vector $\underset{\sim}{x}^{\text{aug}}$ (Definition 0.9), with b as the coefficient on its first entry.

Further reading

- Zhou [14], sections 6 ("Linear functions and operators"; see Definition 0.4) and 7 ("Affine functions"), proves that every linear function is a matrix-vector product by the same route as Theorem 0.4, and that every affine function has the form $\mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$. It defines "affine" differently, by $f(\alpha \underset{\sim}{x} + \beta \underset{\sim}{y}) = \alpha f(\underset{\sim}{x}) + \beta f(\underset{\sim}{y})$ whenever $\alpha + \beta = 1$, and shows that this condition holds exactly for the maps $\mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$, so the two definitions agree.
- Boyd and Vandenberghe [15], chapter 2 ("Linear functions"), covers the vector-to-number case: linear functions as inner products, affine functions, the first-order Taylor approximation as an affine function, and the regression model. Section 8.1 ("Linear and affine functions") covers the vector-to-vector case. A free PDF is on the book's website.

Definition 0.10 (Hyperplane)**.** A **hyperplane** [16] in $\mathbb{R}^p$ is the set of points $\underset{\sim}{x} \in \mathbb{R}^p$ satisfying

$$\underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0$$

for some vector $\underset{\sim}{w} \in \mathbb{R}^p \setminus \{\underset{\sim}{0}\}$ and number $b \in \mathbb{R}$. The number b is the offset of the hyperplane.

Definition 0.11 (Normal vector of a hyperplane)**.** In Definition 0.10, the vector $\underset{\sim}{w}$ is a **normal vector** [17] of the hyperplane.

Example 0.11 (A line in $\mathbb{R}^2$, and why the normal vector must be nonzero)**.**

- With $\underset{\sim}{w} = (1, 1)$ and $b = -1$, the hyperplane in $\mathbb{R}^2$ is the line $x_1 + x_2 - 1 = 0$. The point $(1, 0)$ is on it, since $1+0-1 = 0$; the point $(1, 1)$ is not, since $1+1-1 = 1 > 0$, so it lies in the half-space where $\underset{\sim}{w} \cdot \underset{\sim}{x} + b > 0$.

- With $\underset{\sim}{w} = \underset{\sim}{0}$, the equation reads $b = 0$: if $b \neq 0$ no point satisfies it, and if $b = 0$ every point does, so the set is empty or all of $\mathbb{R}^2$, not a line. That degenerate case is why the definition requires $\underset{\sim}{w} \neq \underset{\sim}{0}$.

Exercise 0.19 (Testing whether a vertical line is a hyperplane)**.** A vertical line in $\mathbb{R}^2$ is parallel to the x_2-axis. It cannot be written as $x_2 = mx_1 + k$ for any numbers m and k. Let L be the set of points $\underset{\sim}{x} \in \mathbb{R}^2$ with $x_1 = 3$.
Is L a hyperplane in $\mathbb{R}^2$?

Hint

Look for $\underset{\sim}{w} \neq \underset{\sim}{0}$ and b in Definition 0.10.

Solution

Solution 0.19. Yes. Take $\underset{\sim}{w} = (1, 0)^\top \neq \underset{\sim}{0}$ and $b = -3$. Then

$$\underset{\sim}{w} \cdot \underset{\sim}{x} + b = x_1 - 3,$$

which is 0 exactly when $x_1 = 3$. So L is the hyperplane with these $\underset{\sim}{w}$ and b.

The course website's version of this page has an interactive plot here.

Exercise 0.20 (Write a rule in matrix form)**.** A clinic scores a patient's risk from three measurements.

symbol	meaning	units
x_1	age	years
x_2	systolic blood pressure	mmHg
x_3	body mass index	kg/m^2

The rule in use is

$$\text{score} = 0.4\,x_1 + 0.2\,x_2 + 1.5\,x_3 - 30$$

Write this rule in the form of Equation 1, naming $\underset{\sim}{w}$, b and p explicitly.

Solution

Solution 0.20. Read the coefficients straight off the rule. There are three inputs, so $p = 3$:

$$\underset{\sim}{w} = \begin{bmatrix} 0.4 \\ 0.2 \\ 1.5 \end{bmatrix}, \qquad b = -30$$

and the rule is $f(\underset{\sim}{x}) = \underset{\sim}{w}^\top \underset{\sim}{x} + b$.

Exercise 0.21 (Score a patient)**.** Give the score from the rule in Exercise 0.20 for a patient aged 50 with blood pressure 130 and BMI 28.

Solution

Solution 0.21. For the patient,

$$\underset{\sim}{x} = \begin{bmatrix} 50 \\ 130 \\ 28 \end{bmatrix}$$

$$\begin{aligned} \underset{\sim}{w}^\top \underset{\sim}{x} &= (0.4)(50) + (0.2)(130) + (1.5)(28) \\ &= 20 + 26 + 42 \\ &= 88 \end{aligned}$$

$$\begin{aligned} f(\underset{\sim}{x}) &= 88 - 30 \\ &= 58 \end{aligned}$$

Two things are worth noticing. The coefficients are not comparable as they stand. The weight on BMI, 1.5, is 7.5 times the weight on blood pressure, 0.2, but the contribution of BMI, 42, is less than twice the contribution of blood pressure, 26, because the BMI value, 28, is much smaller than the blood pressure value, 130. So the weight on a feature says nothing on its own about how much that feature contributes to the score. And $b = -30$ changes the score: without it, the score would be 88 instead of 58.

Exercise 0.22 (Which of these are affine in $\underset{\sim}{w}$?)**.** Each of the five expressions is a function of the weights $\underset{\sim}{w} \in \mathbb{R}^p$, with the data $\underset{\sim}{x}$ and y held fixed. Say which ones are affine (Definition 0.7), that is, have the form Equation 1 — an inner product plus an offset — and which of those are also linear (Definition 0.3).

1. $\underset{\sim}{w}^\top \underset{\sim}{x}$
2. $\underset{\sim}{w}^\top \underset{\sim}{x} - y$
3. $(\underset{\sim}{w}^\top \underset{\sim}{x} - y)^2$
4. $\|\underset{\sim}{w}\|_2^2$
5. $w_1 x_1 + 3$

Solution

Solution 0.22.

1. **Affine, and linear.** It is Equation 1 with $b = 0$, which is a linear map by Theorem 0.2: the matrix $\underset{\sim}{x}^\top$ times $\underset{\sim}{w}$.
2. **Affine.** Still an inner product plus an offset; here the offset is $b = -y$, which is a constant because y is held fixed. It is linear only when $y = 0$: otherwise $\underset{\sim}{w} = \underset{\sim}{0}$ gives $-y \neq 0$, and a linear map sends $\underset{\sim}{0}$ to 0 (Theorem 0.1).
3. **Not affine.** Squaring is not an inner product plus an offset. This expression is the squared error of one observation, and its curvature in $\underset{\sim}{w}$ is exactly why fitting a model is an optimization problem rather than a lookup.
4. **Not affine.** By Equation 2 in Vectors this is $\underset{\sim}{w}^\top \underset{\sim}{w}$, which has $\underset{\sim}{w}$ in both slots at once. Hold either slot fixed and it is linear in the other; as a function of the single variable $\underset{\sim}{w}$ it is quadratic.
5. **Affine, not linear.** It is Equation 1 with coefficient vector $\underset{\sim}{a} = (x_1, 0, \ldots, 0)^\top$ and $b = 3$; at $\underset{\sim}{w} = \underset{\sim}{0}$ it equals $3 \neq 0$, so it is not linear (Theorem 0.1). The data $\underset{\sim}{x}$ is untouched — $\underset{\sim}{a}$ is a new vector built from its first entry.

Items 1, 2 and 5 are affine and items 3 and 4 are not, which is the split that matters: a *model* can be affine in its weights while the *quantity we minimize* is not. Squared error and the norm penalty are both curved in $\underset{\sim}{w}$, and that curvature is what makes fitting an optimization problem at all: there is always a downhill direction to follow. Curvature alone does not promise a *single* answer. The squared error of one observation is completely flat along a whole line of weight vectors, every one of which fits that observation exactly, and pinning down one of them is part of what the norm penalty is added for in week 3.

Transposes of sums and products

Theorem 0.1 (Transpose of a sum)**.** *For $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$:*

$$(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$$

In particular, for column vectors $\underset{\sim}{x}$ and $\underset{\sim}{y}$ of the same length:

$$(\underset{\sim}{x} + \underset{\sim}{y})^\top = \underset{\sim}{x}^\top + \underset{\sim}{y}^\top$$

Proof

Proof. Entry (i, j) of each side:

$$\begin{aligned}
\left[(\mathbf{A}+\mathbf{B})^\top\right]_{ij} &= (\mathbf{A}+\mathbf{B})_{ji} && \text{(definition of the transpose)} \\
&= a_{ji} + b_{ji} && \text{(definition of matrix addition)} \\
&= (\mathbf{A}^\top)_{ij} + (\mathbf{B}^\top)_{ij} && \text{(definition of the transpose)} \\
&= \left[\mathbf{A}^\top + \mathbf{B}^\top\right]_{ij} && \text{(definition of matrix addition)}
\end{aligned}$$

□

Theorem 0.2 (Transpose of a product)**.** *For an $m \times k$ matrix $\mathbf{A}$ and a $k \times n$ matrix $\mathbf{B}$:*

$$\underbrace{(\mathbf{AB})^\top}_{n\times m} = \underbrace{\mathbf{B}^\top}_{n\times k}\ \underbrace{\mathbf{A}^\top}_{k\times m}$$

Proof

Proof. Entry (i, j) of each side:

$$\begin{aligned}
\left[(\mathbf{AB})^\top\right]_{ij} &= (\mathbf{AB})_{ji} && \text{(definition of the transpose)} \\
&= \sum_{s=1}^{k} a_{js}\, b_{si} && \text{(definition of matrix multiplication)} \\
&= \sum_{s=1}^{k} (\mathbf{B}^\top)_{is}\, (\mathbf{A}^\top)_{sj} && \text{(definition of the transpose; reorder each product)} \\
&= \left[\mathbf{B}^\top\, \mathbf{A}^\top\right]_{ij} && \text{(definition of matrix multiplication)}
\end{aligned}$$

□

Remark (The order of the factors reverses)

Remark 0.1 (The order of the factors reverses). Transposing a product reverses the order of the factors. Keeping the original order usually gives an undefined product or a different matrix. For example, with $\mathbf{A} = [1,\ 2]$ (1×2) and $\mathbf{B} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ (2×1),

$$\begin{aligned}
\mathbf{AB} &= 1 \cdot 3 + 2 \cdot 4 \\
&= 11,
\end{aligned}$$

so $(\mathbf{AB})^\top = 11$, and

$$\mathbf{B}^\top \mathbf{A}^\top = [3,\ 4] \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 11, \qquad \mathbf{A}^\top \mathbf{B}^\top = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [3,\ 4] = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}.$$

If instead $\mathbf{A}$ is 2×3 and $\mathbf{B}$ is 3×1, then $\mathbf{B}^\top \mathbf{A}^\top$ is a $(1 \times 3)(3 \times 2)$ product, but $\mathbf{A}^\top \mathbf{B}^\top$ is a $(3 \times 2)(1 \times 3)$ product, which is undefined.

Exercise 0.1 (The order of the factors in a transposed square product)**.** Remark 0.1 shows that, for factors that are not square, keeping the original order of the factors in the transpose of a product can give an undefined product. For two square matrices of the same size, both $\mathbf{A}^\top\mathbf{B}^\top$ and $\mathbf{B}^\top\mathbf{A}^\top$ are defined, so the shapes alone cannot settle which order gives $(\mathbf{AB})^\top$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Does $\mathbf{A}^\top\mathbf{B}^\top$ equal $(\mathbf{AB})^\top$ for these $\mathbf{A}$ and $\mathbf{B}$?

Hint

Compute $\mathbf{AB}$ with Definition 0.1 and take its transpose, then compute $\mathbf{A}^\top\mathbf{B}^\top$ and compare the two entry by entry.

Solution

Solution 0.1. No: the two matrices differ, so keeping the order of the factors gives the wrong transpose.
The product $\mathbf{AB}$ and its transpose are

$$\begin{aligned} \mathbf{AB} &= \begin{bmatrix} 1\cdot 1 + 2\cdot 1 & 1\cdot 0 + 2\cdot 1 \\ 0\cdot 1 + 1\cdot 1 & 0\cdot 0 + 1\cdot 1 \end{bmatrix} && \text{(Definition 0.1)} \\ &= \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}, && \text{(multiply and add)} \end{aligned}$$

so $(\mathbf{AB})^\top = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$.

The transposes are $\mathbf{A}^\top = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ and $\mathbf{B}^\top = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, so

$$\begin{aligned} \mathbf{A}^\top\mathbf{B}^\top &= \begin{bmatrix} 1\cdot 1 + 0\cdot 0 & 1\cdot 1 + 0\cdot 1 \\ 2\cdot 1 + 1\cdot 0 & 2\cdot 1 + 1\cdot 1 \end{bmatrix} && \text{(Definition 0.1)} \\ &= \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}. && \text{(multiply and add)} \end{aligned}$$

Entry $(1,1)$ is 3 in $(\mathbf{AB})^\top$ and 1 in $\mathbf{A}^\top\mathbf{B}^\top$. In the reversed order, as Theorem 0.2 says,

$\mathbf{B}^\top\mathbf{A}^\top = \begin{bmatrix} 1\cdot 1+1\cdot 2 & 1\cdot 0+1\cdot 1 \\ 0\cdot 1+1\cdot 2 & 0\cdot 0+1\cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, which matches $(\mathbf{AB})^\top$.

Theorem 0.3 (A 1×1 product equals its own transpose)**.** *If $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are vectors of length p and $\mathbf{M}$ is a $p\times p$ matrix, then $\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v}$ is a number, and*

$$\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v} = \underset{\sim}{v}^\top\mathbf{M}^\top\underset{\sim}{u}.$$

In particular, $\underset{\sim}{u}^\top\underset{\sim}{v} = \underset{\sim}{v}^\top\underset{\sim}{u}$.

Proof

Proof. The product $\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v}$ is a $(1\times p)(p\times p)(p\times 1)$ product, so it is a 1×1 matrix. The transpose of a 1×1 matrix is itself, so

$$\begin{aligned} \underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v} &= (\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v})^\top && \text{(a } 1\times 1 \text{ matrix equals its transpose)} \\ &= \underset{\sim}{v}^\top\mathbf{M}^\top(\underset{\sim}{u}^\top)^\top && \text{(transpose of a product, twice)} \\ &= \underset{\sim}{v}^\top\mathbf{M}^\top\underset{\sim}{u} && \text{(transposing twice changes nothing)} \end{aligned}$$

The transpose step is Theorem 0.2. Taking $\mathbf{M}=\mathbf{I}$ gives the last statement. □

Example 0.1 (Two ways to write the same number)**.** Take $\underset{\sim}{u}=(1,2)$, $\underset{\sim}{v}=(3,-1)$ and $\mathbf{M}=\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Then $\mathbf{M}\underset{\sim}{v}=(1,-3)$, so

$$\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v} = 1\cdot 1+2\cdot(-3) = -5.$$

Also $\mathbf{M}^\top=\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ and $\mathbf{M}^\top\underset{\sim}{u}=(1,8)$, so

$$\underset{\sim}{v}^\top\mathbf{M}^\top\underset{\sim}{u} = 3\cdot 1+(-1)\cdot 8 = -5.$$

The two products are equal, as the theorem says. In least squares, this is why the two cross terms of $(\mathbf{X}\underset{\sim}{\beta}-\underset{\sim}{y})^\top(\mathbf{X}\underset{\sim}{\beta}-\underset{\sim}{y})$ are equal and can be combined.

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